October University for Modern Sciences and Arts Faculty of Management Sciences MGT_420II Computer-Based Information System Graduation Project II

Logical Design for the Integration of Acetone-Level Monitoring with Continuous Glucose Sensors for Enhanced Diabetes Management

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Executive Summary

Diabetes is regarded as one of the most challenging health issues in Egypt and the world, as millions of people are affected by the complications resulting from unreliable blood sugar levels and late diagnosis of metabolic diseases. The serious complication of DKA (diabetic ketoacidosis) can cause excessive amounts of ketones to be produced by the body due to a lack of insulin, and is a potentially life-threatening condition. While the current Continuous Glucose Monitoring (CGM) systems have made diabetes management easier, they offer limited monitoring capabilities as a lack of early detection of the metabolic changes associated with ketones.

This graduation project presents a new integrated healthcare monitoring system which integrates continuous glucose monitoring (CGM) with real-time acetone detection to obtain a more comprehensive picture of the patient's metabolism. The proposed system consists of a set of wearable biosensors, an ESP32 microcontroller, cloud computing technologies, and a mobile application that continuously gathers, processes and analyzes the physiological data. The system will measure both glucose and acetone levels at the same time, thus detecting early signs of ketosis and DKA before they develop to their severe state. The setup will automatically alert the patients and healthcare providers with abnormal readings to allow timely intervention in healthcare.

Moreover, the medical information gathered is securely stored on the cloud and doctors can monitor patients remotely, access the patient's medical history, and examine trends and make clinical decisions remotely. The solution also enables data storage offline so that the data can be monitored without interruption even if the internet connection is lost. The proposed project will have a positive impact on patient safety, a positive effect on reducing emergency admissions to hospital, and a positive impact on improving the quality of diabetes management particularly for those who live in remote and under-served areas.

Moreover, the system enables Egypt's digital healthcare revolution by integrating Internet of Medical Things (IoMT) technologies with smart medical monitoring. The project will provide a dual-biomarker monitoring approach to change diabetes management from reactive to proactive diabetes prevention and early intervention.

Business Case

2.1. Project Overview

The healthcare sector is currently experiencing rapid digital transformation through the adoption of smart medical technologies, wearable devices, cloud computing, and Internet of Medical Things (IoMT) solutions. One of the most critical healthcare challenges driving this transformation is diabetes mellitus, which continues to increase at alarming rates worldwide and especially in Egypt. According to the International Diabetes Federation (IDF), Egypt currently has approximately 13.2 million adults living with diabetes, placing the country among the top nations globally in diabetes prevalence (International Diabetes Federation, 2024).

Despite the widespread use of Continuous Glucose Monitoring (CGM) systems in diabetes management, existing systems still suffer from a major medical limitation because they only monitor glucose levels without detecting ketone-related metabolic changes. This creates a serious clinical gap in detecting diabetic ketoacidosis (DKA), which is one of the most dangerous and life-threatening diabetes complications. DKA develops when insulin deficiency causes excessive ketone production, leading to metabolic acidosis that may rapidly result in hospitalization, organ failure, or death if not detected early. Scientific studies confirmed that acetone levels strongly correlate with ketone production and can act as an early non-invasive biomarker for metabolic deterioration and ketoacidosis risk (Saasa et al., 2018; Wang & Wang, 2013).

The proposed graduation project aims to upgrade traditional CGM systems by integrating continuous acetone monitoring alongside glucose monitoring within a single wearable healthcare solution. The system combines biosensors, ESP32 microcontroller technology, cloud computing, mobile healthcare applications, and real-time analytics to create a smart monitoring platform capable of detecting early metabolic abnormalities before severe symptoms appear.

The importance of this project lies in shifting diabetes care from reactive emergency treatment toward proactive preventive healthcare. Instead of waiting for dangerous symptoms to appear, the system continuously analyzes glucose and acetone trends simultaneously and alerts patients and healthcare providers immediately when abnormal readings are detected. This contributes to reducing emergency hospital admissions, improving patient safety, supporting remote healthcare services, and enhancing healthcare accessibility for patients living in rural or underserved regions.

Furthermore, the project directly supports modern healthcare domain objectives including smart healthcare systems, remote patient monitoring, IoMT integration, preventive medicine, and data-driven clinical decision-making. The system also provides secure cloud-based medical records that enable long-term health analysis and support future medical research regarding diabetes behavior and metabolic patterns.

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2.2. Project Scope in Terms of Opportunities/Problem

The proposed project's scope is designed to tackle key challenges in current Continuous Glucose Monitoring (CGM) systems and enhance the quality of diabetes healthcare monitoring. While current CGM technologies give the patient continuous glucose information to help them manage their diabetes, the technology does not fully give the patient an understanding of the patient's metabolic state as these current technologies don't detect if there is a change in the blood that may lead to diabetic ketoacidosis (DKA). DKA is one of the most serious and life-threatening complications of diabetes and can develop quickly if not caught early. Research studies have demonstrated that glucose monitoring is not enough to detect early metabolic deterioration as ketone production might rise before severe glucose abnormalities are evident (Saasa et al., 2018). Thus, treating only by blood sugar monitoring alone may delay proper medical treatment and lead to hospitalization and major complications.

Currently, one of the most significant health challenges covered by the project is the lack of continuous monitoring of acetone in the current wearable health care systems. Acetone has a close relationship with the production of ketones and is a good early indicator of diabetic ketoacidosis and metabolic instability. It was found through scientific studies that the concentration of acetone increases significantly during the state of ketosis and is directly proportional to the concentration of blood ketones, so that the use of acetone monitoring will be very useful for the early detection of ketosis and for preventive health care intervention (Rydosz, 2018).

The majority of current ketone testing techniques are, however, invasive, labor-intensive, and not connected to real-time wearable monitoring systems, thus there is a significant gap in contemporary diabetes management solutions.

Medical systems still don't have intelligent real-time data analysis to detect abnormal metabolic patterns before the patient enters a critical stage, apart from monitoring limitations. Current monitoring equipment frequently offer individual data or limited analytical tools that do not have advanced warning systems or automation to assist with early intervention in healthcare. This limitation undermines the effectiveness of remote health monitoring, and limits the ability to use preventive healthcare strategies over and against emergency medical treatment.

The proposed project aims to overcome these issues by developing an integrated smart healthcare system with three key functional requirements: continuous glucose monitoring, continuous acetone monitoring, and intelligent healthcare data analysis. Wearable biosensors are connected to an ESP32 microcontroller and a healthcare platform in the cloud, which continuously monitor glucose and acetone levels. All the collected data is processed as it comes in and dangerous metabolic trends are detected, generating instant alert notifications when abnormal conditions are detected. In addition, the system securely stores patient data in the cloud, enabling healthcare professionals to track patient health remotely, analyze historical patient data to identify medical trends, and make better clinical decisions.

The project helps to improve patient safety, decrease diabetic complications, increase remote healthcare access, and promote the digitalization of the healthcare sector using the Internet of Medical Things (IoMT) technologies via these features. The system also enables opportunities for proactive healthcare management moving diabetes treatment from reactive “emergency” to continuous “preventive” monitoring and early intervention.

References

Saasa, V., Malwela, T., Beukes, M., Mokgotho, M., Liu, C. P., & Mwakikunga, B. (2018). *Sensing technologies for detection of acetone in human breath for diabetes diagnosis and monitoring*. Diagnostics, 8(1), 12. https://www.mdpi.com/2075-4418/8/1/12

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International Diabetes Federation. (2024). *Egypt diabetes report*. IDF Diabetes Atlas. https://diabetesatlas.org/data-by-location/country/egypt/

2.3. Target Beneficiaries

The proposed healthcare system is designed for multiple groups of beneficiaries in the healthcare domain, as it is considered to facilitate diabetes monitoring, early detection of metabolic risk and remote healthcare services.

Diabetes Patients

The main target beneficiaries are diabetes patients, particularly those with Type 1 Diabetes and insulin-dependent Type 2 Diabetes which are at increased risk of diabetic ketoacidosis (DKA). The patients should be monitored continuously as the metabolic deterioration may occur very quickly in the early stages without any symptoms. The system could benefit patients by

offering real-time glucose and acetone monitoring and early warning alerts for safer disease management and to minimize emergency complications.

Healthcare Providers

Physicians, endocrinologists, diabetes experts, nurses and emergency healthcare personnel have access to real-time patient data and long-term metabolic trends. The system enhances clinical decision making, enabling healthcare providers to keep track of patients remotely, detect abnormal patterns early, and intervene when problems start to emerge.

Patients

Patients in hospitals and medical facilities. There is a decrease in emergency admissions for diabetic ketoacidosis and severe glucose instability in hospitals and clinics. Moreover, the technology enables healthcare digitalization initiatives with cloud-based monitoring, eHealthcare integration, and remote patient management in line with current healthcare trends of the IoMT.

Families and Caregivers

Family members and caregivers can receive alerts and notifications regarding abnormal patient conditions, which helps them respond quickly in emergency situations and can ultimately increase patient safety, especially children and elderly diabetic patients.

Medical Researchers, Public Health Organizations

Long-term glucose and acetone data can be stored in the cloud to provide valuable healthcare records and contribute to future medical research, epidemiology studies, predictive healthcare analytics, and national diabetes management programs.

References

International Diabetes Federation. (2024). *Diabetes in Egypt*. International Diabetes Federation. https://idf.org/our-network/regions-and-members/middle-east-and-north-africa/members/egypt/

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2.4. Expected Demand Size in Egypt/Region

Current diabetes prevalence statistics and the increasing demand for sophisticated remote healthcare monitoring technologies support the projected demand for the proposed system. The projected demand for the proposed system is backed by the current diabetes prevalence statistics and the escalating demand for advanced remote healthcare monitoring technologies.

Based on the *International Diabetes Federation (IDF) 2024* numbers, there are currently ~ 13.2 million adults with diabetes in Egypt, accounting for nearly 22.4% of all adults. The number will grow to about 24.7 million adults by 2050.

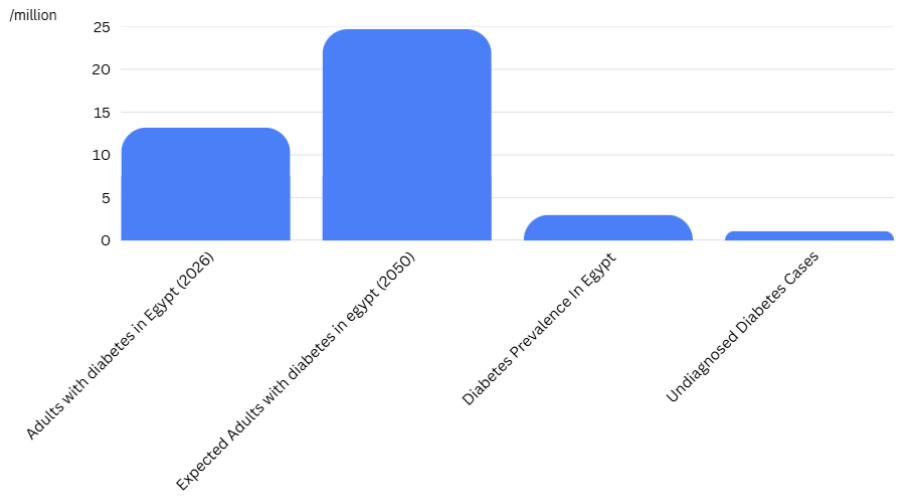
Moreover, the IDF says there were around 62% undiagnosed diabetes cases in Egypt, suggesting that a significant number of people face the possibility of complications if it remains untreated and lack medical intervention.

This generates a high demand in the healthcare sector for systems that can:

- Continuous remote monitoring

- Early detection of metabolic risk.
- Real-time healthcare alerts
- Preventive healthcare management
- Healthcare services based on smart clouds.

This is even more important when the patient is a diabetic dependent patient or a patient with high risk for developing diabetic keto-acidosis.



This Figure Shows the number of registered cases of adult diabetes in Egypt amounting to (13.2m) people in 2026 and expected the numbers to grow to (24.7m) people by 2050. It also takes into account the number of unregistered diabetes cases which are also included in the demand size.

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Features of the System

3.1. Theoretical Background

Diabetes mellitus is a chronic metabolic disease with abnormal blood glucose regulation due to inadequate insulin secretion or utilization of insulin by the body. Diabetic patients need continuous monitoring as unstable glucose levels can cause severe complications like hypoglycemia, hyperglycemia, cardiovascular diseases, nerve damage, kidney failure, and diabetic ketoacidosis (DKA). Typical blood glucose monitoring approaches rely primarily on intermittent finger prick tests, which are not capable of continuous real-time monitoring and can miss rapid changes in metabolism between checks.

To enhance diabetes management, a new class of wearable health care technologies, Continuous Glucose Monitoring (CGM) systems, were introduced. A new class of wearable health care technologies that measure the glucose level continuously throughout the day, Continuous Glucose Monitoring (CGM) systems, were introduced to improve diabetes management. They have a range of features that help patients monitor their glucose levels more accurately, including real-time readings, trend analysis, and alerts for abnormal glucose levels, which can help to lower diabetes-related complications. Yet, although these benefits exist, there remains one big drawback to traditional CGM devices: they are based on glucose monitoring and do not monitor other metabolic indicators that are important for DKA.

Diabetic ketoacidosis (DKA) is a serious condition that occurs when the blood sugar drops too low, forcing the body to use fat as fuel that causes an increased

production of ketones. Acetone is one of the major ketone related biomarkers and has been extensively studied as a non-invasive marker of metabolic instability and increased risk for DKA. Studies of research showed that there is a high correlation between acetone concentration and the production of blood ketones, which is very useful for early diagnostic and health prevention measures (Rydosz 2018). So, the incorporation of acetone monitoring with the existing glucose monitoring can be very helpful in detecting metabolic abnormalities early and improve the diabetes management system.

The proposed system is based on concept of smart healthcare and Internet of Medical Things (IoMT), in which the wearable biosensors, embedded system, cloud computing and mobile health application are combined together in order to provide continuous healthcare monitoring. The system consists of biosensors that are connected to an ESP32 microcontroller to collect glucose and acetone readings in real-time. The collected data is sent to a cloud-based healthcare platform where analytical algorithms analyze the data, detect any abnormal patterns, and send automated notifications to the patient and the health care provider. It enhances the ability to deliver preventive care, remote patient monitoring, and real-time clinical decision-making, while decreasing reliance on hospital-based monitoring systems.

References

World Health Organization. (2023). *Diabetes*. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/diabetes

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3.2. Comparative Analysis

There are currently several Diabetes monitoring systems available in the healthcare market such as the Dexcom G7, FreeStyle Libre 3, Medtronic Guardian and Eversense E3 systems, which are all wearable Continuous Glucose Monitoring (CGM) systems. These systems have greatly helped improve the management of diabetes by offering real-time glucose monitoring, trend visualization, and alert notifications, assisting patients in monitoring blood glucose levels better than with finger-prick methods. New healthcare research shows that CGM technologies can provide continuous glucose monitoring every few minutes and help to improve glycemic control and patient decision-making.

The Dexcom G7 system is part of the most advanced commercially available CGM system and offers continuous real-time monitoring, smartphone compatibility, automatic alerts and insulin delivery system compatibility. Likewise, the FreeStyle Libre 3 system offers real-time glucose monitoring, a long sensor life and mobile application support. Medtronic Guardian systems also provide predictive glucose alerts and also permit integration with insulin pumps for automated diabetic issues management. These systems, however, may be very sophisticated in their glucose monitoring capabilities, but have limited monitoring of acetone or ketone functions.

A major drawback of current systems is the lack of continuous monitoring of ketone related biomarker for diabetic ketoacidosis (DKA). Existing ketone monitoring approaches are predominantly based on blood

or urine ketone test strips which are invasive, labor-intensive, cumbersome, and independent of any ongoing wearable monitoring system. Medical research has confirmed that ketone monitoring is a critical part of monitoring insulin deficiency, and early signs of ketosis development, particularly in patients with Type 1 diabetes and those who are at high risk of developing DKA (diabetic ketoacidosis). But there are still emerging ketone monitoring technologies that are not commonly used in commercial CGM devices.

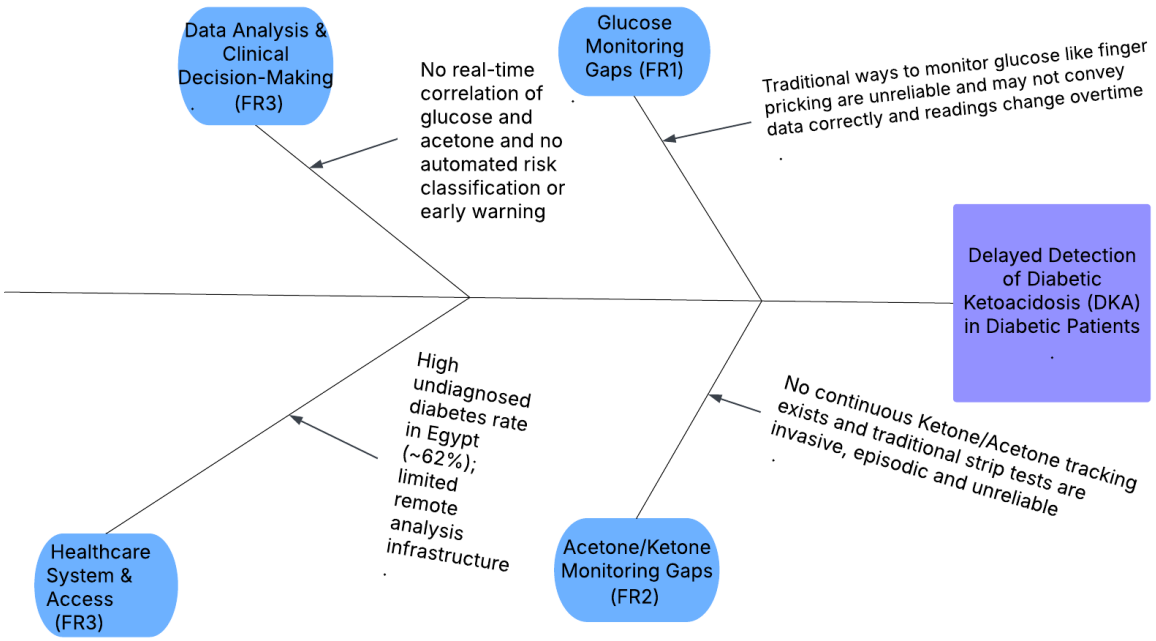
This system is also different from the existing healthcare monitoring systems as it combines glucose monitoring and acetone monitoring in a single smart wearable system. The proposed project will measure both glucose and acetone levels, unlike traditional CGM systems which only measure glucose, to give a more complete picture of the patient's metabolic condition. Because acetone concentration is tightly linked to the production of ketones and risk of DKA, adding acetone monitoring enables the system to identify potentially dangerous metabolic changes at an earlier stage than a standard CGM system. This is a significant healthcare benefit for preventing diabetes and intervening early in an emergency situation.

The other key difference is the incorporation of intelligent healthcare data analysis in the proposed system. The current CGM systems primarily offer numerical glucose values and graphing, whereas the proposed system would analyze both the glucose data and the acetone data in real time, provide visual interpretation of the trends, and notify healthcare providers of abnormal metabolic patterns. The system also features cloud computing and remote healthcare monitoring capabilities, enabling healthcare professionals to access patient information remotely, track long-term patient trends and assist with clinical decisions using cloud-based medical systems. A comparison between existing healthcare monitoring system and proposed system is summarized in following table:

System	Glucose Monitoring	Acetone Monitoring	Real Time Alerts	Data Analysis	Cloud Integration	Early DKA Detection
Dexcom G7	yes	no	yes	Limited	yes	Limited
Freestyle Libre 3	yes	no	yes	Limited	yes	Limited
Medtronic Guardian	yes	no	yes	Moderate	yes	Limited
Traditional Ketone Testing	no	yes	no	no	no	yes
Proposed system	yes	yes	yes	Advanced	yes	yes

Thus, the project proposed presents an innovative healthcare monitoring approach, which integrates dual-biomarker monitoring, intelligent healthcare analytics, cloud-based healthcare services and preventive medical intervention in a single smart healthcare system. The integration addresses several critical challenges in current CGM technologies and helps enhance patient safety, minimize diabetic complications, and aid in the digitization of healthcare systems.

3.3 Fishbone Analysis



3.4. Functional Requirements

The proposed healthcare monitoring system is based on three key functional requirements that facilitate ongoing diabetes management, early detection of metabolic risk and intelligent healthcare monitoring.

FR1: Monitor Glucose Levels

The system will use wearable biosensors equipped with ESP32 microcontroller to continuously measure the level of glucose. All accumulated glucose data will be sent to the health care platform and shown in the mobile app interface. The system will also log glucose data to a cloud database for later monitoring and health analysis. Further, the system shall be capable of issuing alarms when glucose levels are outside of safe ranges to minimize the danger of hypoglycemia and hyperglycemia.

FR2: Monitor Acetone Levels.

The system shall continuously sense the level of acetone as a good indicator of the production of ketones and risk of diabetic ketoacidosis. Acetone measurements will be taken by integrated biosensors and be analyzed with glucose values in order to gain a more complete picture of the patient's metabolic state. The system shall be able to identify abnormal levels of acetone and shall provide early warnings if hazardous metabolic patterns are detected. This function is designed to enhance early diagnosis and preventive healthcare interventions, at the point of avoidance of severe complications.

FR3: Analyze Healthcare Data

The system shall continuously process glucose and acetone data in real time using analytical algorithms to detect abnormal trends and dangerous metabolic changes. The analysis process shall identify potential diabetic ketoacidosis risks and support intelligent healthcare decision-making by generating continuous monitoring reports for patients and healthcare providers. In addition, the system shall store the analyzed healthcare information in the cloud system for long-term analysis, future review, patient history tracking, and remote healthcare services.

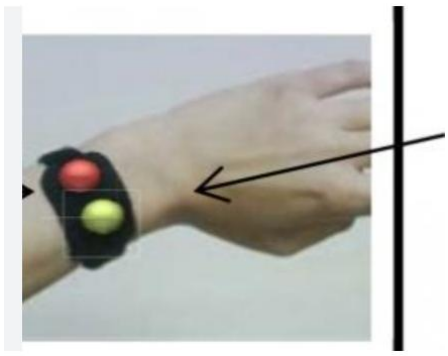
3.5 Technology Needed

There are a number of technologies essential for the system to function properly. The first requirement of the system is a continuous glucose monitoring device, which will be able to record glucose values throughout the day. The technology of CGM is already widely adopted for the treatment of diabetes and they provide real-time readings (Sha et al., 2021). An acetone skin sensor is the second thing needed in the system.

This is for the detection of acetone in exhaled skin odor: some research results suggested there was a relationship between acetone production and acetone in the skin caused by raised temperature. Modern acetone sensors are typically based on nanometers or

nanowires, which enhance the sensor's sensitivity and accuracy (Liang & Wang, 2024).

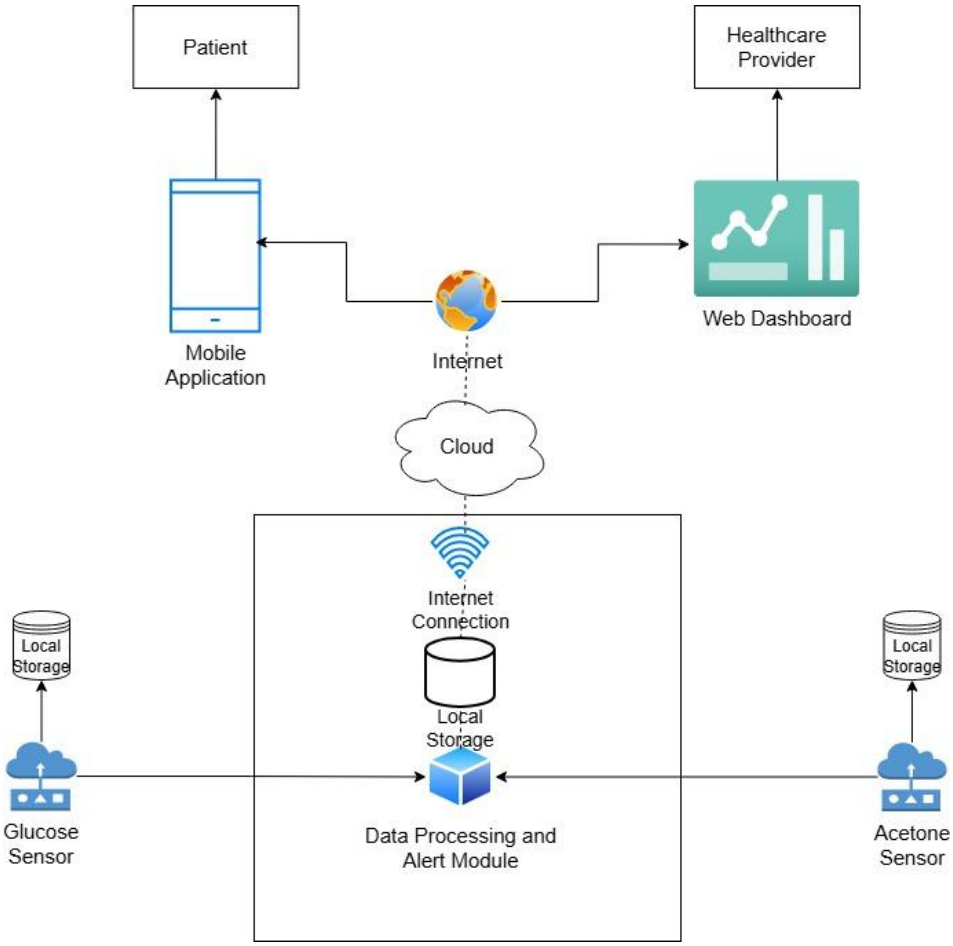
Additionally, a microcontroller is required which will be able to gather data and upload it to the cloud storage, and which will be able to communicate with the mobile application, via Wi-Fi or Bluetooth. When readings are abnormal or alarming, the mobile phone application needs to display the readings to the patient, and it must also have a reliable, robust cloud storage with doctors/caregivers access to the cloud storage to reach patient's data.

CGM	
Acetone Skin Sensor	
EP32 Microcontroller	

Application	
Cloud Storage	

3.6 IT Infrastructure

The IT infrastructure must support all the following: Continuous data flow, remote access and secure storage. IoT architecture will be used in the system where sensors send data to the cloud. This is consistent with modern IoMT (Internet of Medical Things) trends that support real-time monitoring (Villegas-Ch et al., 2023).



• **Ali Nagy Part in FRs**

FR Name	Monitor Glucose Levels
Description	The system will continuously measure the level of glucose using the wearable biosensors connected to the ESP32 microcontroller, the collected glucose reading are then transformed real-time to the cloud-based healthcare platform in order to either store, or conduct analysis or generate alerts, the system also shows the current

	readings of glucose levels as well as historical trends on the patient's mobile application interface.
What	Continuously monitor and measure the concentration levels of glucose in the blood through the wearable biosensors and transmit the collected readings to the cloud-based healthcare monitoring platform for storage, analysis in real-time, trend visualization, and automated alert generation when abnormal levels of glucose are detected.
Why	Glucose is the main biomarker for managing diabetes, continuous monitoring of glucose levels allows the detection of any dangerous conditions early on, such as hypoglycemia (critically low glucose) and hyperglycemia (critically high glucose) before they ascend into life-threatening medical emergencies, without real-time monitoring of glucose levels, patients and healthcare providers cannot take well-timed informed medical decisions, this increases the risk of diabetes complications and increases the risk of emergency hospitalizations, this system is made to address the delay of the traditional finger-prick testing method.
Where	The monitoring process starts at the weared biosensor that is attached to the patient's body, which in turn captures glucose

	levels continuously, sensor readings are recieved and processed by the ESP32 microcontroller locally before transmitting them by Wi-Fi or Bluetooth to the cloud healthcare program, the data of the glucose (readings and current levels), are then stored on the patient's application's database and the healthcare providers' on the mobile application interface, they can be accessed through the doctor's web dashboard.
When	The system operates continuously in the 24 hours of the day, whenever the wearable device of monitoring is active and connected to the platform of the healthcare, glucose readings are captured in regular time intervals throughout the entire day, if internet connection is not available, the data is stored locally on the device and get automatically synced to the cloud once connectivity is restored, alerts fire immediately when glucose levels changes from the predetermined safe glucose levels.
Dependency	Depends on the glucose biosensor, ESP32 microcontroller, Wi-Fi/Bluetooth connectivity model, cloud database, mobile healthcare application, and real-time data analysis module (FR3).
Microservices	Glucose Sensor Service, Data Collection Service, Cloud Storage Service, Real-Time Analytics Service, Alert Notification

	Service, Mobile Application Service.

• **Nour Marai Part in FRs**

FR Name	Monitor Acetone Levels
Description	The system continuously monitors acetone levels using wearable biosensors to detect ketone production and identify early signs of diabetic ketoacidosis (DKA). The collected acetone readings are transmitted to the healthcare platform for analysis and alert generation.
What	Measure acetone concentration levels continuously and send the readings to the cloud-based healthcare monitoring system for storage, analysis, and alert processing.
Why	Acetone is strongly associated with ketone production and diabetic ketoacidosis. Monitoring acetone levels helps detect metabolic instability early, reduces emergency complications, and supports preventive healthcare intervention before the patient reaches a critical condition.
Where	The monitoring process occurs through wearable biosensors connected to the ESP32 microcontroller, while data processing and storage occur within the cloud healthcare platform and mobile application.
When	The system operates continuously in real time whenever the wearable monitoring device is

	active and connected to the healthcare platform. Alerts are generated immediately when abnormal acetone levels are detected.
Dependency	Depends on the acetone biosensor, ESP32 microcontroller, cloud database, internet connectivity, mobile healthcare application, and real-time data analysis module.
Microservices	Acetone Sensor Service, Data Collection Service, Cloud Storage Service, Real-Time Analytics Service, Alert Notification Service, Mobile Application Service

•Youssef Alaa Part in FRs

FR Name	Analyze Healthcare Data
Description	The system analyzes glucose and acetone readings in real time to identify abnormal trends, dangerous glucose levels, and potential risk of diabetic ketoacidosis. It also provides healthcare alerts, monitoring reports, and stores information in the cloud for future analysis and remote healthcare monitoring
What	Analyze glucose and acetone readings in real time, detect abnormal health patterns, and present the analysis results to the patient and healthcare providers.
Why	Absence of intelligent healthcare data analysis.

	To solve the issue of delayed diagnosis of metabolic abnormalities and support early healthcare decision-making.
Where	In the Cloud-based Healthcare Platform, Patient Mobile Application, and Doctor Web Dashboard.
When	When new glucose and acetone readings are received, the data is analyzed continuously in real time.
Dependency	(FR01) and (FR02)
Microservices	Healthcare Data Analysis Microservice

Quality Map	
Have we built the right product?	Yes. The proposed healthcare monitoring system addresses a real and significant healthcare problem related to diabetes management and early diabetic ketoacidosis (DKA) detection. Through the project scope, business problem analysis, and 4Ws framework, the project clearly identifies the target users, healthcare environment, healthcare objectives, and the importance of integrating continuous glucose monitoring with wearable acetone skin biosensor technology. The functional requirements further validate that the proposed solution directly supports real-time healthcare monitoring, healthcare alert generation, and healthcare

Have we built the product right?	data analysis required for diabetic healthcare management. Yes. The project foundation was developed systematically through clearly defined project objectives, functional requirements, healthcare monitoring scope, and supporting scientific references. The business problem analysis, related healthcare technologies, wearable biosensor concepts, cloud healthcare architecture, and healthcare monitoring functionalities are consistently documented and logically connected. Furthermore, the functional requirements are structured in a way that supports future system analysis, UML modeling, and healthcare process documentation throughout the project lifecycle.

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4. Business Processes

4.1. Data Flow Diagram (DFD)

4.1.1. Data Flow Diagram (DFD) Context Level

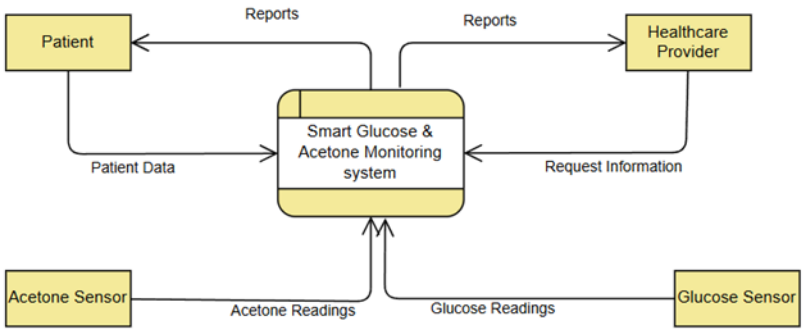


Figure 4.1 (Developed by Students)

The context-level Data Flow Diagram presents a high-level overview of the Integrated Glucose and Acetone Monitoring System and shows how the system interacts with external entities. At this level, the entire system is represented as a single process in order to define the system boundary and illustrate the flow of data without showing internal operations.

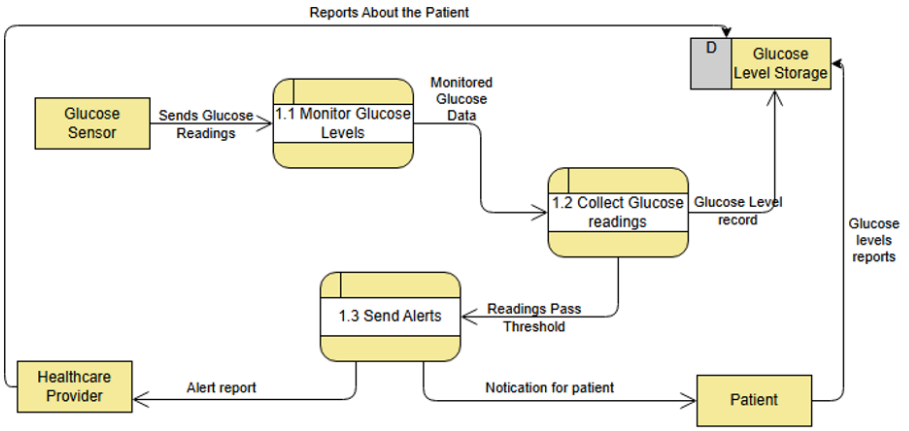
The system mainly interacts with three external entities: the Patient, the Healthcare Provider, and the Cloud System. The patient sends glucose and acetone readings to the system through wearable sensors and provides login information when accessing the application. In return, the system sends alerts, notifications, and health reports that help the patient monitor their condition and respond early to abnormal health situations.

Healthcare providers access the system by submitting login credentials and data requests. The system provides them with patient alerts, health dashboards, and historical medical data, allowing doctors and caregivers to monitor patients remotely and make timely medical decisions.

The cloud system acts as an external service responsible for storing medical data and supporting data analysis. The monitoring system uploads patient records and audit logs to the cloud, while the cloud returns stored records and processed information when needed

4.1.2. Data Flow Diagram (DFD) 0 Level

• Ali Nagy Part in (DFD) 0 Level



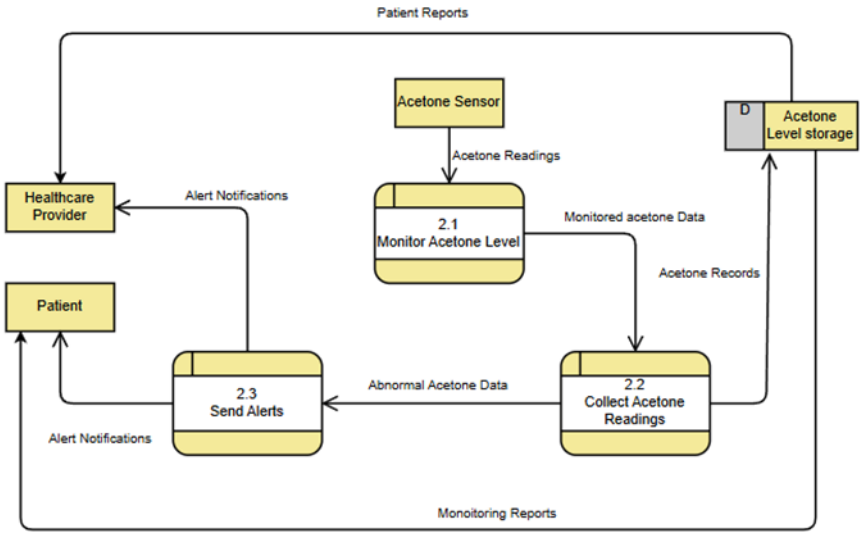
Documentation of Data Flow Diagram (DFD) 0 level for FR1

Explanation of FR1 lvl 0 DFD: The Whole flow of data first starts with the glucose sensor that is attached and connected to the diabetic patient, the glucose sensor continuously monitors the glucose levels inside the patient's blood by seeing it's components and their percentages, the sensor then starts to collect the data/level of glucose level in the blood of the diabetic patient (it stores them in it's local storage as a backup while simultaneously comparing the readings),

the glucose sensor then compares the level of readings it's currently capturing with the pre-set caution level, the glucose sensor if it captures a reading breaking the threshold, will send an alert to the patient as a notification on their mobile phone through the application,

an alert is also sent to their healthcare provider through their own health dashboard in order to keep track with the patient and be notified with critical and imminent dangers, to also decide whether medical intervention is needed or not.

- **Nour Marai Part in (DFD) 0 Level**



Documentation of Data Flow Diagram (DFD) 0 level for FR2

In FR2, a 0-Level DFD is shown to represent the internal healthcare monitoring process for the task of monitoring acetone concentration levels via wearable acetone biosensors that are integrated into the healthcare monitoring system. The DFD shows how the processes in the system interact with each other, the data elements they operate on, and the external entities they interact with, in terms of the flow of acetone readings, healthcare monitoring data, healthcare warning notifications, and monitoring reports.

The Acetone Sensor external entity is constantly sending the concentration of acetone to Process 2.1 “Monitor Acetone Levels”. This process carries out the continuous healthcare monitoring operation and

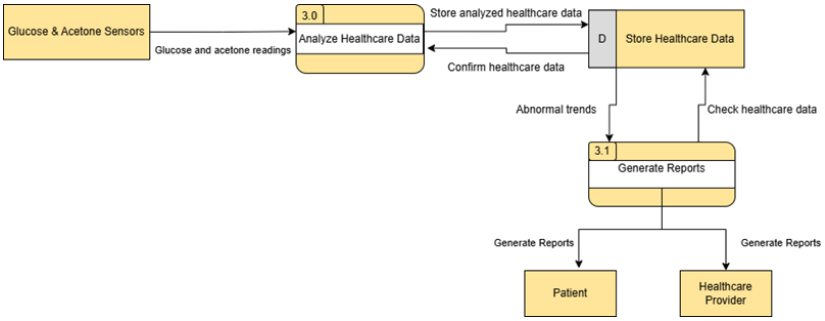
sends the data it monitors to Process 2.2 “Collect Acetone Readings”.

Process 2.2 “Collect Acetone Readings” refers to the collection and sharing of acetone concentration readings in the healthcare monitoring system. The gathered acetone data is saved in the Acetone Level Storage data store for healthcare monitoring and reporting purposes. Furthermore, if there is some abnormal concentration in the acetone data, abnormal acetone data is sent to Process 2.3 “Send Alerts”.

The healthcare notification function of healthcare monitoring system is represented as Process 2.3 (Send Alerts). This process creates and disseminates healthcare warning messages to the external entities Patient and Healthcare-Provider, in the form of warnings for abnormal concentration levels of Acetone. Moreover, monitoring report generation and delivery is supported to enable ongoing healthcare monitoring and medical follow-up operations with the patient.

The Acetone Level Storage data store contains the healthcare monitoring records of the acetone monitoring process, and enables the health healthcare monitoring system to generate patient monitoring reports.

•Youssef Alaa Part in (DFD) 0 Level



Documentation of Data Flow Diagram (DFD) 0 level for FR3

Analyze Healthcare Data

This process is responsible for analyzing of real-time healthcare data from glucose and acetone sensors. The system detects unusual patterns, high glucose levels, and a possible risk for DKA. Analyzed healthcare data is uploaded to the healthcare cloud database to be analyzed for the future and remotely monitored in healthcare.

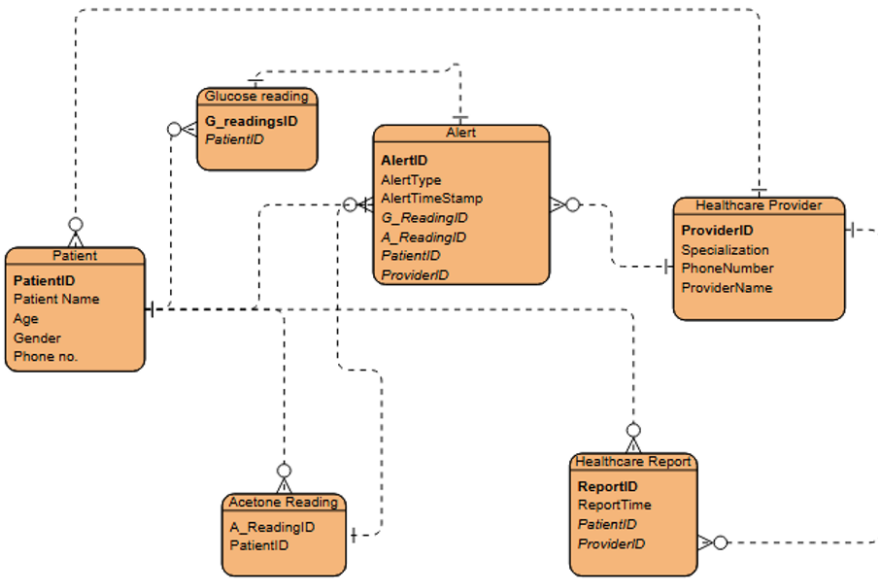
Functionality:

The process starts when glucose and acetone sensor services provide real-time readings to Process 3.0 “Analyze Healthcare Data.” The system analyzes the glucose and acetone readings to detect abnormal trends, dangerous metabolic changes, and potential DKA risk. After the analysis is completed, the analyzed healthcare data is stored in the “Store Healthcare Data” data store.

Then, Process 3.1 “Generate Reports” retrieves the analyzed healthcare data from the data store and generates monitoring reports. These reports are provided to the patient and healthcare provider and include the analysis result, risk level, and recommended action.

4.2. Entity Relationship Diagram (ERD)

4.2.1. Entity Relationship Diagram (ERD) Model



4.2.2. Documentation of Entity Relationship Diagram (ERD)

Patient

Primary Key	PatientID
Foreign Key	—
Attributes	PatientName, Age, Gender, PhoneNo
Patient & Glucose Reading	One patient can have many glucose readings.
Patient & Acetone Reading	One patient can have many acetone readings.
Patient & Alert	One patient can receive many healthcare alerts.
Patient & Healthcare Report	One patient can have many healthcare reports.

Healthcare Provider

Primary Key	ProviderID
Foreign Key	—

Attributes	ProviderName, Specialization, PhoneNumber
Healthcare Provider & Alert	One healthcare provider can receive many healthcare alerts.
Healthcare Provider & Healthcare Report	One healthcare provider can review many healthcare reports.

Glucose Reading

Primary Key	G_ReadingID
Foreign Key	PatientID
Attributes	—
Glucose Reading & Patient	Many glucose readings belong to one patient.
Glucose Reading & Alert	One glucose reading may generate healthcare alerts.

Acetone Reading

Primary Key	A_ReadingID
Foreign Key	PatientID
Attributes	—
Acetone Reading & Patient	Many acetone readings belong to one patient.
Acetone Reading & Alert	One acetone reading may generate healthcare alerts.

Alert

Primary Key	AlertID
Foreign Key	G_ReadingID, A_ReadingID, PatientID, ProviderID
Attributes	AlertType, AlertTimeStamp
Alert & Patient	Many healthcare alerts can belong to one patient.
Alert & Healthcare Provider	Many healthcare alerts can be received by one healthcare provider.

Alert & Glucose Reading	Alerts may be generated from abnormal glucose readings.
Alert & Acetone Reading	Alerts may be generated from abnormal acetone readings.

Healthcare Report

Primary Key	ReportID
Foreign Key	PatientID, ProviderID
Attributes	ReportTime
Healthcare Report & Patient	Many healthcare reports can belong to one patient.
Healthcare Report & Healthcare Provider	Many healthcare reports can be reviewed by one healthcare provider.

5.1. The OOAD Steps and Role of The UML Modeling

Unified Modeling Language (UML) is a normalized model with graphical language that's providing a common visual notation to describe and design software systems. UML provides an accepted format that's unified and internationally allows developers of the system and stakeholders to communicate system requirements and design it clearly and essentially.

In UML, an information system is composed of several key components. First, the functional requirements are described to define and know the needs and assumptions of the users. Second, a use case model is developed to show the services and functionalities needed and purposed from the system. Third, a class diagram model is used to explain and define the system's structural components and the relationships of them. Fourth, sequence diagrams are employed to describe the connections and message that exchange system over time between classes. Finally, activity diagrams are being used for presenting the workflow and operational processes that system need in a flowchart-like manner.

In addition to UML diagrams, the system modeling process is developed by many traditional analysis tools like Data Flow Diagram (DFD) and the Entity Relationship Diagram (ERD), which focus and help in representing the movement of data and database structure separately.

Object-Oriented Design and Analysis (OOAD) is a construction technique used in this project. Representing a system, especially a set of interrelated objects is the main goal of OOAD, a methodical technique that is implemented through several consecutive steps, as explained below:

Investigation: The current phase focuses at how an entirely new system needs to be created or a current one needs to be modified. It also shows researching the effect and viability of the any better suggested modifications.

Analysis: In the analysis stage, system needs are founded and examined. This can be carried out by examining organizational documents, interviewing participants, and researching the existing system.

Logical Design: The logical framework and requirements of particular system are the main things of this phase. UML diagrams are used widely, framework, and functioning of systems without taking into account the specifics of their actual implementation.

Physical Design: The system's technical framework is created during this stage. Databases, networking, programming of the system, and software and hardware technology selection are all included here in this.

Implementation: Systems setup, evaluation, and instruction for users are all part of this step. Once the system is fully implemented inside the company, it undergoes validation using a variety of methods to guarantee correctness and dependability.

Maintenance: This last stage guarantees that the system will continue to function after deployment. It includes fixing mistakes, implementing modifications, and creating improvements to satisfy changing organizational needs.

5.2. The Logical Design

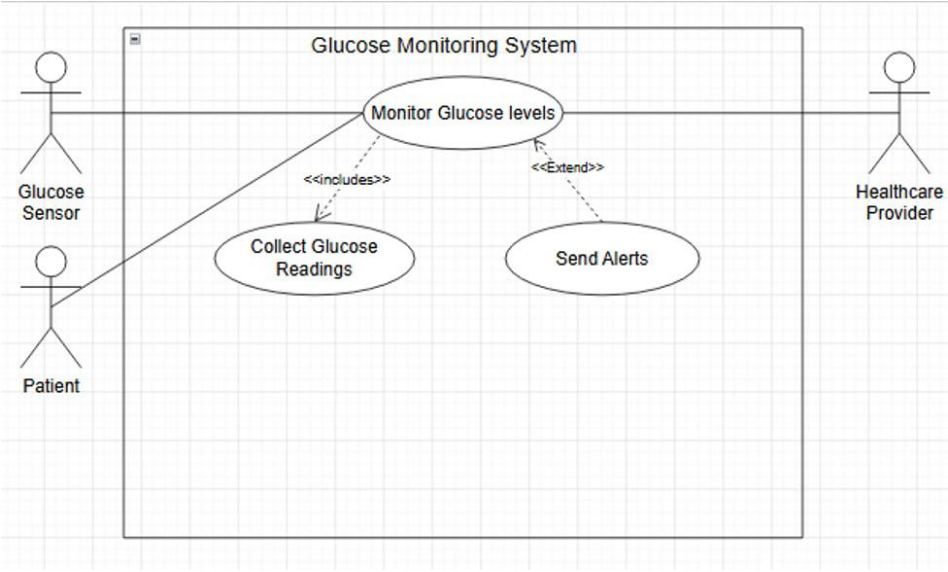
5.3. Use-Case Diagram

5.3.1. Use-Case Model

5.3.2. Documentation of The Use-Case Model

• Ali Nagy Part in Documentation of The Use-Case Model

FR1: Monitor Glucose Levels



Use Case 1: Monitor Glucose levels

Use Case Name	Monitor Glucose Levels
Actor	Patient, Healthcare Provider, Glucose Sensor
Microservices	Glucose Sensor Service, Data Collection Service, Cloud Storage Service, Alert Notification Service, Mobile Application Service
Type	Primary

Cross Reference	FR1
Brief Description	The system will continuously track the patient's blood glucose concentration using a wearable biosensor that's connected to the ESP32 microcontroller, to detect dangerous fluctuations such as hypoglycemia or hyperglycemia. It mandatorily includes Collect Glucose Readings and will conditionally extend to Send Alerts when a critical threshold is passed.
Detailed Description (Flow of Events)	**Basic Flow:** 1. **Monitor Glucose Levels:** the system will receive the levels of glucose and then evaluates them against safe pre-set ranges 2. **Collect Glucose Readings (<<Include>>):** The biosensor will capture and collect the glucose concentration and send it to the Data Collection Service with the patient ID and the timestamp. 3. **Send Alerts (<<Extend>> – Conditionally):** When the reading breaches the safe pre-set range, warning notifications are immediately sent to the

	patient and healthcare provider to notify them. **Expected Outcomes:** Uninterrupted continuous real-time glucose monitoring for both the patient and healthcare provider. **Common Pitfalls:** 1. The Biosensor may malfunction causing missing readings. 2. Loss of connectivity may interrupt data transmissions. 3. Incorrect calibration of thresholds may cause the alarm to falsely go off 4. Cloud service unavailability disrupting storage and losing the data.
Data Types	**Glucose Reading (mg/dL):** Real-time blood glucose value tagged with patient ID and timestamp. Example*: 240 mg/dL, 23759.* **Alert Notification:** a Warning message is generated when a critical threshold is breached.
Pre-condition	1. the Glucose biosensor is attached and transmitting. 2. Patient account is registered in the system. 3. Cloud healthcare database is accessible.

Post-condition	1. Glucose readings are recorded in the cloud database. 2. Dashboards reflect the latest glucose values. 3. Alerts are dispatched immediately if a threshold is breached.

Use Case 1.2: Collect Glucose reading

Use Case Name	Collect Glucose Readings
Actor	Glucose Sensor
Microservices	Glucose Sensor Service, Data Collection Service
Type	Secondary – Include
Cross Reference	FR1
Brief Description	The wearable glucose biosensor will continuously measure the blood glucose concentration and transmits each of the reading to the healthcare monitoring platform by the ESP32 microcontroller.
Detailed Description (Flow of Events)	1. **Capture Reading:** The biosensor measures blood glucose concentration in mg/dL. 2. **Validate Reading:** The ESP32 checks the value for the completeness and integrity. 3. **Transmit to Data Collection Service:** The validated reading is sent with the patient ID, the sensor ID, and the timestamp. **Expected Outcomes:** A continuous stream of accurate glucose readings that will be delivered to the platform. **Troubleshooting:** 1. Confirm that the biosensor is properly attached. 2. Verify ESP32 connection is stable.

	3. Check transmitted values are within a valid range.
Data Types	**Glucose Reading (mg/dL):** Raw blood glucose concentration that is measured by the biosensor. **Patient ID:** Unique identifier linking the reading to a patient. **Timestamp:** Date and time of the captured glucose reading.
Pre-condition	1. Glucose biosensor is powered and attached well to the patient 2. ESP32 is connected to the healthcare monitoring platform.
Post-condition	1. A validated glucose reading will be delivered to the Data Collection Service for storage and threshold evaluation.

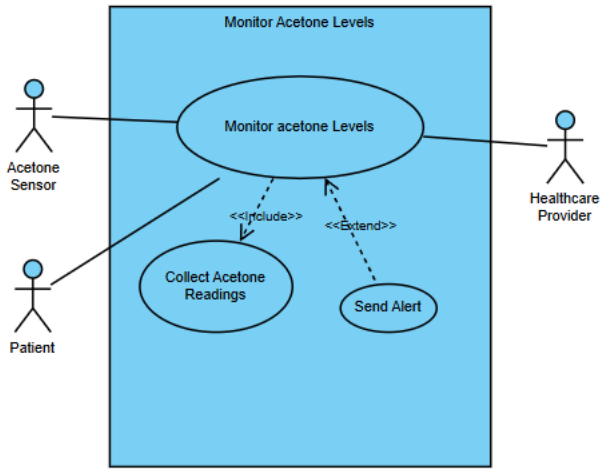
Use Case 3: Send Alerts

Use Case Name	Send Alerts
Actor	Patient, Healthcare Provider
Microservices	Alert Notification Service, Mobile Application Service
Type	Secondary – Extend
Cross Reference	FR1
Brief Description	When a glucose reading breaches the pre-set limit to the sensor, the system will immediately send notifications and warnings about the glucose levels.

Detailed Description (Flow of Events)	1. **Detect Threshold Breach** The platform will identify a glucose reading that is outside of the safe range and classifies it as Warning or Critical. 2. **Generate Alert** The Alert Notification Service will construct a warning message with the glucose value, alert type, patient ID, and timestamp. 3. **Notify Patient** The notification is then pushed to the patient by the mobile healthcare application. 4. **Notify Healthcare Provider** The alert is then sent to the healthcare provider by the healthcare dashboard. 5. **Log Alert** The alert is stored in the cloud database for any future clinical review. **Expected Outcomes:** Both parties are notified in real time whenever a dangerous glucose level is breached. **Possible Errors:** 1. Down-time from the notification systems. 2. Network unavailability resulting in delaying the alert transmission. 3. False alerts that result from incorrect threshold calibration.
Data Types	**Alert Notification:** Warning message with alert type, glucose value, patient ID, and timestamp. **Glucose Reading (mg/dL):** The reading that triggered the alert. *Example: 310 mg/dL.*
Pre-condition	1. the Platform is active and receiving glucose readings.

	2. A reading that has breached the critical threshold. 3. Alert Notification Service is running.
Post-condition	1. Notifications that is delivered to patient and healthcare provider. 2. Alert record that is stored in the cloud database.

- **Nour Marai Part in Documentation of The Use-Case Model**



Use Case Name	**Monitor Acetone Levels**
Actor	Patient Healthcare Provider Acetone Sensor

Microservices	Acetone Sensor Service, Data Collection Service, Cloud Storage Service, Alert Notification Service, Mobile Application Service
Type	Primary
Cross Reference	FR2
Brief Description	A series of wearable acetone biosensors are attached to the ESP32 microcontroller to continuously monitor the concentration of acetone. The acetone readings are sent to the healthcare monitoring platform, stored securely, and continuously monitored, to aid in monitoring a diabetic's health and detecting any elevated levels of acetone. The most common application is to Monitor Acetone Levels and optional extensions would be Send Alerts if an abnormal acetone level is detected.
Detailed Description (Flow of Events)	Acetone Monitoring Process: The wearable acetone sensor continuously monitors the concentration levels of acetone and uploads the concentration data along with patient ID and timestamp to the health care monitoring platform via Acetone Sensor Service. Basic Flow: **1. Monitor Acetone Levels** Acetone is sent to the healthcare monitoring platform through the Acetone Sensor Service, and is continuously monitored. This system is used for validation and monitoring of the real-time value of acetone measured in the area. **2. Collect Acetone Readings** The acetone biosensor continuously captures acetone concentration levels from the patient through wearable skin sensing technology. The ESP32 microcontroller will receive the sensor reading and send it to the Data Collection Service along with the patient ID and timestamp data. **3. Store Acetone Data**

	Secure Storage of Acetone Data The Acetone Data The Cloud Storage Service securely stores the acetone readings and the healthcare monitoring history in the healthcare cloud database. In case of a lost internet connection, the readings will be temporarily stored locally and automatically synced afterwards when the internet connection is restored. Detects Abnormal level of Acetone ,Send Alerts (<> – Conditionally): If abnormality is detected, the Alert Notification Service triggers alerts and warning notifications and instantly sends them to the patient and health care provider via the mobile health care app and health care dashboard. **Expected Outcomes:** The system can continuously record and store the acetone value properly and enable continuous monitoring of the health care, and can give warning messages when the value of acetone is abnormal. **Common Pitfalls:** • Sensor failure or unstable acetone readings. • Internet connectivity problems can lead to failure in data synchronization. • Cloud-based healthcare storage service can fail temporarily. • Alert service may be down temporarily resulting in failure in alert notifications.
Data Types	**Acetone Reading (ppm):** The current acetone concentration value received from the Acetone Sensor Service and stored with timestamp and patient ID. *"Example: 1.8 ppm."* **Acetone Monitoring Report:** Structured monitoring information containing acetone value, timestamp, patient ID, monitoring status, and alert information.

	Alert Notification: Warning notification generated whenever abnormal acetone levels are detected.
Pre-condition	A Acetone Sensor Service has been started and linked to a Healthcare Monitoring Platform.A Acetone Sensor Service has been running and linked to a Healthcare Monitoring Platform. Patient's account is activated in healthcare monitoring system. Cloud storage solutions for healthcare are available. The acetone biosensor and the ESP32 microcontroller is working properly.
Post-condition	The healthcare cloud-based database can successfully retrieve and store acetone readings. Reports are constantly updated. When abnormal acetone levels are found, alerts are immediately sent to the patient and health care provider. The healthcare dashboard allows healthcare providers to remotely view the most up-to-date acetone monitoring.

Use Case Name	**Collect Acetone Readings**
Actor	Acetone Sensor
Microservices	Acetone Sensor Service, Data Collection Service
Type	Secondary – Include
Cross Reference	FR2
Brief Description	The wearable acetone biosensor continuously captures acetone concentration levels from the patient's skin and transmits the readings to the healthcare monitoring platform through the ESP32 microcontroller.
Detailed Description (Flow of Events)	1. **Capture Acetone Readings** The first step is to capture Acetone Reading with the acetone biosensor which is used to measure the concentration level of the acetone from a patient's skin. 2. **Validate sensor reading**

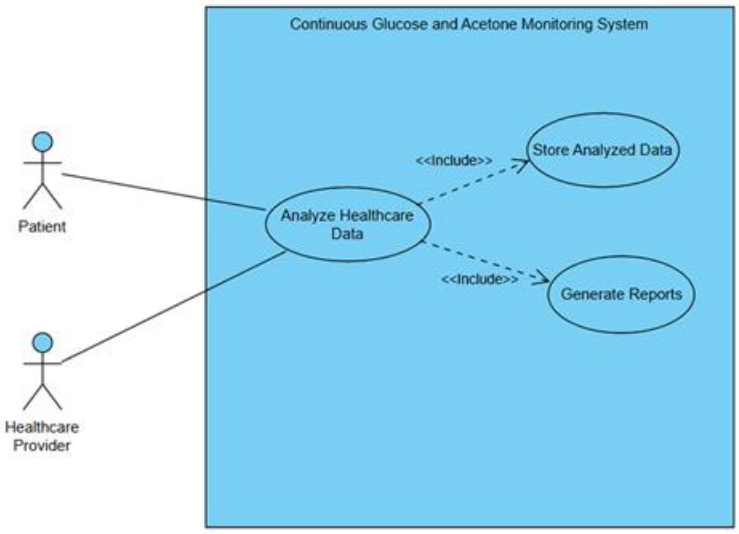
	The ESP32 microcontroller validates sensor reading before sending it out. 3. **Send reading to data collection** The validated reading is sent to the Data Collection Service with the following data: timestamp, sensor ID, and patient ID. **Expected Outcomes:** Acetone readings are transmitted to the healthcare monitoring platform in real time with accuracy. Troubleshooting: Take the following measures in case of errors: • Check that the sensor is stable. • Make sure that communication between the ESP32 and the sensor is working properly. • Ensure that the sensor data is not missing or corrupted.
Data Types	**Acetone Reading (ppm):** Breath acetone concentration value generated by the wearable biosensor. **Timestamp:** Date and time of the captured sensor reading. **Patient ID:** Unique identifier associated with the patient.
Pre-condition	The acetone biosensor and ESP32 microcontroller are active and connected to the healthcare monitoring platform.
Post-condition	Acetone readings are transmitted successfully to the Data Collection Service for healthcare monitoring and storage.

Use Case Name	**Send Alerts**
Actor	Patient Healthcare Provider

Microservices	Alert Notification Service, Mobile Application Service
Type	Secondary - Extend
Cross Reference	FR2
Brief Description	The system automatically notifies the patient and the health care provider in the event of abnormal levels of acetone were detected during monitoring.
Detailed Description (Flow of Events)	1. **Detect Abnormal Acetone Levels** The healthcare monitoring platform detects abnormal acetone concentration levels received from the monitoring process. 2. **Generate Alert Notification** The Alert Notification Service generates a warning notification containing the acetone value, timestamp, patient ID, and alert status. 3. **Send Alert to Patient** The warning notification is sent instantly to the patient through the mobile healthcare application. 4. **Send Alert to Healthcare Provider** The healthcare provider receives the warning notification through the healthcare dashboard for remote monitoring and medical follow-up. 5. **Store Alert Record** The generated alert information is stored in the healthcare cloud database for future monitoring and medical review. **Expected Outcomes:** Warning notifications are delivered successfully to both the patient and healthcare provider whenever abnormal acetone levels are detected.

	Possible Errors: • Failure in alert notification delivery. • Internet connectivity issues. • Temporary downtime in the Alert Notification Service.
Data Types	**Alert Notification:** Warning message generated whenever abnormal acetone levels are detected. **Acetone Reading (ppm):** Current acetone concentration value associated with the alert. **Timestamp:** Date and time associated with the alert generation process.
Pre-condition	The health monitoring platform is alive and working and is getting readings of acetone. The Alert Notification Service is running.
Post-condition	Alert notifications are sent successfully to the patient and healthcare provider. Alert records are stored in the healthcare cloud database for future healthcare monitoring and review.

• Youssef Alaa Part in Documentation of The Use-Case Model



Use Case Name	**Analyze Healthcare Data**
Actor	Patient Healthcare Provider
Microservices	Healthcare Data Analysis Microservice
Type	Primary
Cross Reference	FR01 – Monitor Glucose Levels FR02 – Monitor Acetone Levels
Brief Description	This use case describes how the Healthcare Data Analysis Microservice analyzes real-time glucose data from FR1 and acetone data from FR2. The microservice detects abnormal metabolic trends using analytical algorithms. The Analyze Healthcare Data use case includes Store Analyzed Data and Generate Reports.

Detailed Description (Flow of Events)	**Healthcare Data Reading:** Glucose readings measured in mg/dL and acetone readings measured in ppm are received from FR1 and FR2. Each reading is linked with a patient ID and timestamp. The Healthcare Data Analysis Microservice analyzes both readings together using dual-biomarker analysis to determine the patient's metabolic risk level, which can be Normal, Warning, or Critical. **Basic Flow:** **1.Analyze Healthcare Data** The Healthcare Data Analysis Microservice receives glucose and acetone readings from FR1 and FR2. It compares the paired values with safe ranges and patient historical data to classify the metabolic risk level. **2.Store Analyzed Data** After the analysis is completed, the result of each analysis cycle is stored in the cloud healthcare database. The stored data includes patient ID, glucose value, acetone value, timestamp, and risk level. If the cloud connection is not available, the data is stored locally and synchronized later when the connection is restored. **3.Generate Reports** The system generates a healthcare monitoring report based on the analyzed glucose and acetone data. The report includes glucose value, acetone value, timestamp, risk level, and recommended action. **Expected Outcomes:** The Healthcare Data Analysis Microservice processes incoming glucose and acetone readings, calculates the metabolic risk level, stores the analyzed data in the cloud database, and generates

	monitoring reports for follow-up by the patient and healthcare provider. **Alternative / Exception Flows** **Incomplete analysis may happen if glucose or acetone data from FR1 or FR2 is delayed or missing.** Cloud database connection failure may happen, so the system stores data locally until the connection is restored. Wrong classification may happen if there is an algorithm error.
Data Types	**Glucose Reading (mg/dL)** The current blood glucose data is received as input from the Glucose Sensor Service (FR1) and stored with a timestamp and patient ID. Example: 240 mg/dL. **Acetone Reading (ppm)** The acetone concentration data received from the Acetone Sensor Service or FR2. Example: 1.8 ppm. **Metabolic Risk Report** A structured output generated after each analysis cycle. It includes risk level, glucose value, acetone value, timestamp, and recommended action.
Pre-condition	FR1 (Glucose Sensor Service) is ON and sends readings. FR2 (Acetone Sensor Service) runs and sends readings. Patient accounts are registered and connected to the monitoring system. Data can be stored on the cloud healthcare platform.

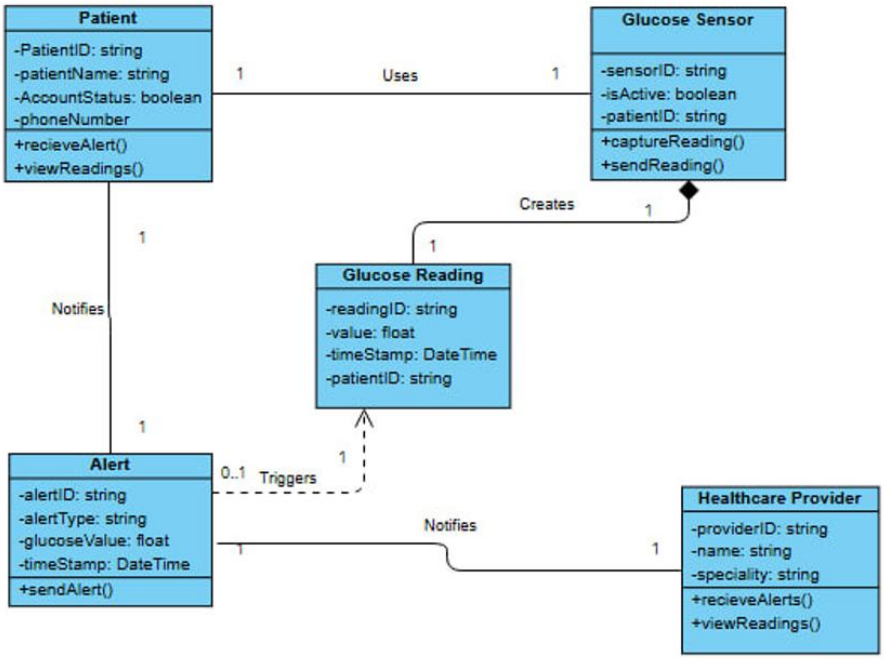
Post-condition	The healthcare data is analyzed and the risk level is calculated. The analyzed data is stored in the cloud database or temporarily stored locally if there is no connection. A monitoring report is generated based on the analysis result. If abnormal patterns are detected, the report includes the appropriate risk status and recommended action.

5.4. Classes Diagram

5.4.1. Class Diagram Model Version 18

5.4.2. Documentation of The Class Model

• Ali Nagy Part in Documentation of The Class Model



Class Name	Patient
Description	This class here represents the diabetic patient that is wearing the glucose sensor and receives warning and readings through their mobile application.

Attributes	PatientID, PatientName, AccountStatus, PhoneNumber
Associations	**GlucoseSensor** **Alert :** The patient is linked to the glucose sensor and it's attached to them and it then continuously collects their glucose readings and send alerts to them when the threshold is passed in glucose readings
Operations	ReceiveAlert(), ViewReadings()

Class Name	GlucoseSensor
Description	Represents the wearable glucose biosensor that is attached to the patient's body and is responsible for continuously capturing the blood glucose concentration levels and transmitting them to the healthcare monitoring platform.
Attributes	SensorID, IsActive, PatientID
Associations	**Patient** **GlucoseReading** : The sensor is used by the patient and it continuously creates glucose readings for the monitoring system to track.
Operations	CaptureReading(), SendReading()

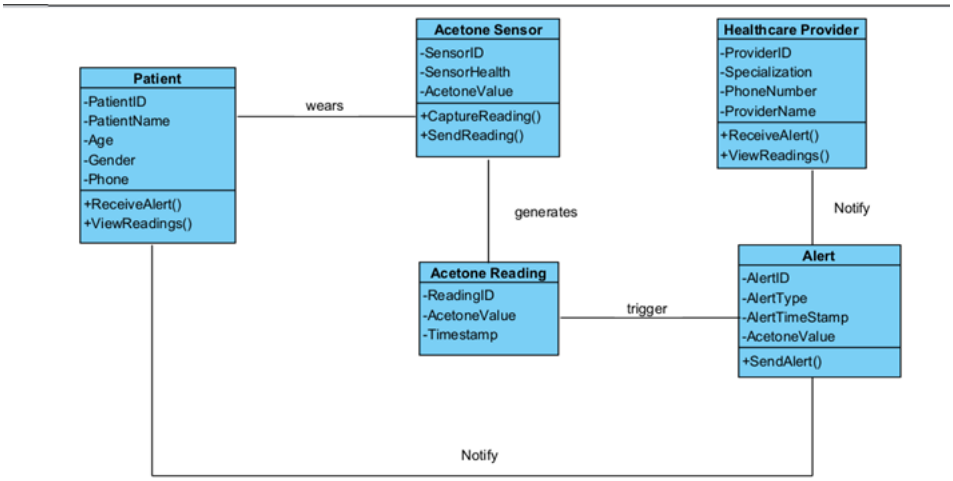
Class Name	GlucoseReading
Description	This Represents the blood glucose concentration reading that is captured by the wearable biosensor and then transmitted to the healthcare monitoring platform for evaluation.
Attributes	ReadingID, Value, TimeStamp, PatientID
Associations	**GlucoseSensor** **Alert** **:** The reading is created by the glucose sensor and it triggers an alert when the

	glucose value exceeds the pre-defined/pre-set critical threshold.
Operations	-

Class Name	Alert
Description	Represents a warning notification generated whenever a glucose reading breaches the predefined critical threshold during the monitoring process.
Attributes	AlertID, AlertType, GlucoseValue, TimeStamp
Associations	**GlucoseReading** **Patient** **HealthcareProvider:** The alert is triggered by an abnormal glucose reading and warning are sent as notifications to both the patient and healthcare provider.
Operations	SendAlert()

Class Name	HealthcareProvider
Description	This class Represents the healthcare provider who is responsible for remotely monitoring the glucose levels of the patient and receiving any warning notifications when abnormal readings are detected.
Attributes	ProviderID, Name, Speciality
Associations	**Alert** **:** The healthcare provider receives an alert notification through the healthcare dashboard whenever any abnormal glucose levels are detected.
Operations	ReceiveAlerts(), ViewReadings()

- **Nour Marai Part in Documentation of The Class Model**



Class Name	**Patient**
Description	Represents the patient who uses the healthcare monitoring system to monitor acetone concentration levels and receive healthcare alerts.
Attributes	PatientID, PatientName, Age, Gender, PhoneNumber
Associations	*Acetone Sensor* *Acetone Reading* *Alert* The patient receives healthcare alerts and monitoring readings from the monitoring platform.
Operations	ReceiveAlert(), ViewReadings()

Class Name	**Acetone Sensor**
Description	Represents the wearable acetone biosensor responsible for capturing acetone concentration levels from the patient's skin and sending the readings to the monitoring system.

Attributes	SensorID, SensorStatus, AcetoneValue
Associations	*Patient* *Acetone Reading* The sensor captures acetone readings and generates monitoring data for the healthcare system.
Operations	CaptureReading(), SendReading()

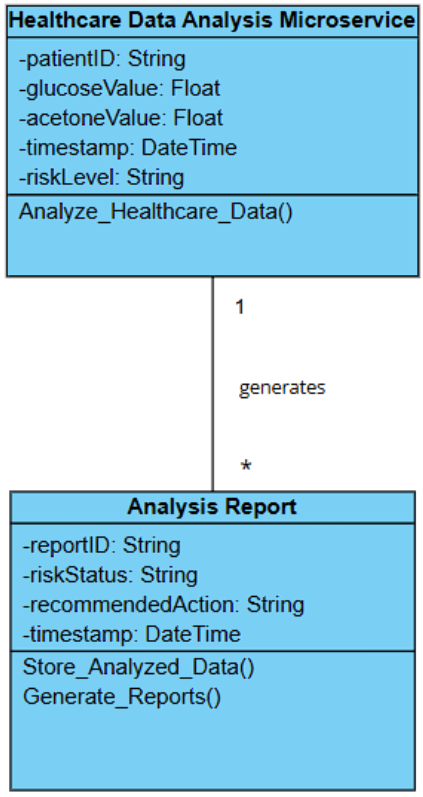
Class Name	**Acetone Reading**
Description	Represents the acetone concentration reading generated from the wearable acetone biosensor and transmitted to the healthcare monitoring system.
Attributes	ReadingID, AcetoneValue, TimeStamp
Associations	*Acetone Sensor* *Alert* The acetone reading is stored in the healthcare platform and used to generate healthcare alerts when abnormal readings are detected.
Operations	

Class Name	**Alert**
Description	Represents healthcare warning notifications generated whenever abnormal acetone concentration levels are detected during the monitoring process.

Attributes	AlertID, AlertType, AlertTimeStamp, AcetoneValue
Associations	*Acetone Reading* *Patient* *Healthcare Provider* The alert class sends healthcare warning notifications to both the patient and healthcare provider.
Operations	SendAlert()

Class Name	**Healthcare Provider**
Description	Represents the healthcare provider responsible for remotely monitoring patient acetone readings and receiving healthcare warning notifications.
Attributes	ProviderID, ProviderName, Specialization, PhoneNumber
Associations	*Alert* The healthcare provider receives healthcare alerts and views patient monitoring records remotely through the healthcare dashboard.
Operations	ReceiveAlert(), ViewReadings()

• Youssef Alaa Part in Documentation of The Class Model



Class: Healthcare Data Analysis Microservice
Attributes
patientID: A unique identifier assigned to each patient. It is used to connect the incoming glucose and acetone readings with the correct patient record in the system. **glucoseValue:** The blood glucose reading measured in mg/dL and received from the Glucose Sensor Service or FR1 in real time. It is used as the main biomarker for diabetes monitoring. **acetoneValue:**

The acetone concentration measured in ppm and received from the Acetone Sensor Service or FR2. It is used with glucose to identify abnormal metabolic patterns and possible DKA risk.

timestamp:
The date and time when the glucose and acetone readings are received. It helps in tracking patient history and analyzing trends over time.

riskLevel:
The result of the analysis process. It shows the metabolic risk level of the patient, such as Normal, Warning, or Critical.

Operations

Analyze_Healthcare_Data():
This operation analyzes the paired glucose and acetone readings using dual-biomarker analysis. It compares the readings with safe ranges and patient history to determine the patient's metabolic risk level. After the analysis is completed, the result is sent to the Analysis Report class to store the analyzed data and generate reports.

Class: Analysis Report

Attributes

reportID:
A unique identifier for each generated analysis report. It helps in tracking each report in the cloud database.

riskStatus:
The risk status generated from the analysis result, such as Normal, Warning, or Critical.

recommendedAction:

A recommended action based on the patient's risk status. For example, if the risk is Warning or Critical, the report can recommend medical follow-up or urgent attention. **timestamp:** The date and time when the analysis report is created. It is used for history tracking and audit purposes.
Operations
Store_Analyzed_Data(): This operation stores the final analysis result in the cloud healthcare database, including patient ID, glucose value, acetone value, risk level, risk status, recommended action, and timestamp. **Generate_Reports():** This operation generates healthcare monitoring reports based on the analyzed data. These reports can be used later by the patient and the healthcare provider for remote monitoring and follow-up.

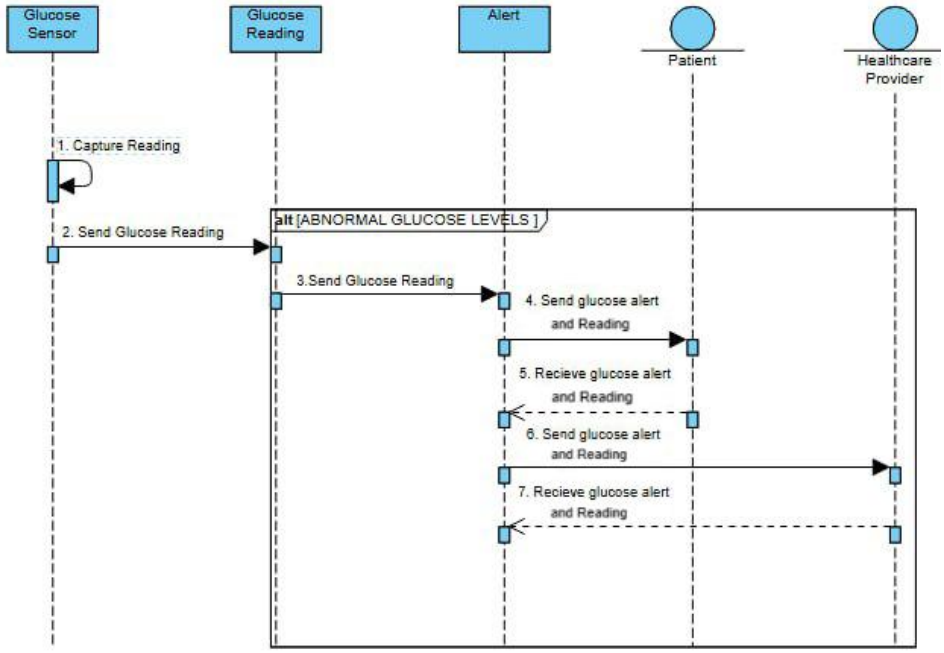
Quality Map	
Have we built the right product?	Yes. The developed system models accurately represent the healthcare functionalities required to support continuous glucose monitoring, acetone monitoring, healthcare data analysis, healthcare alert generation, and healthcare reporting within the proposed smart healthcare monitoring system. The use case diagrams, DFDs, ERD, and class diagrams collectively demonstrate that the system design fulfills the healthcare objectives and operational requirements

	identified during the project analysis phase. The healthcare workflows and healthcare monitoring interactions remain aligned with the proposed diabetic healthcare monitoring solution.
Have we built the product right?	Yes. The system analysis and logical design models were developed with strong consistency and traceability between the functional requirements and system processes. The use case diagrams are directly reflected in the DFD processes, while the ERD accurately represents the persistent healthcare monitoring data required by the healthcare monitoring workflows. In addition, the class diagrams maintain consistency with the use case operations, entities, and healthcare functionalities. The relationships, operations, entities, data flows, and healthcare monitoring processes are logically connected and systematically documented without isolated components or disconnected functionalities.

5.5. Sequence Diagram

5.5.1. Sequence Diagram Models

- **Ali Nagy Part in Documentation of The Sequence Model**



Explanation:

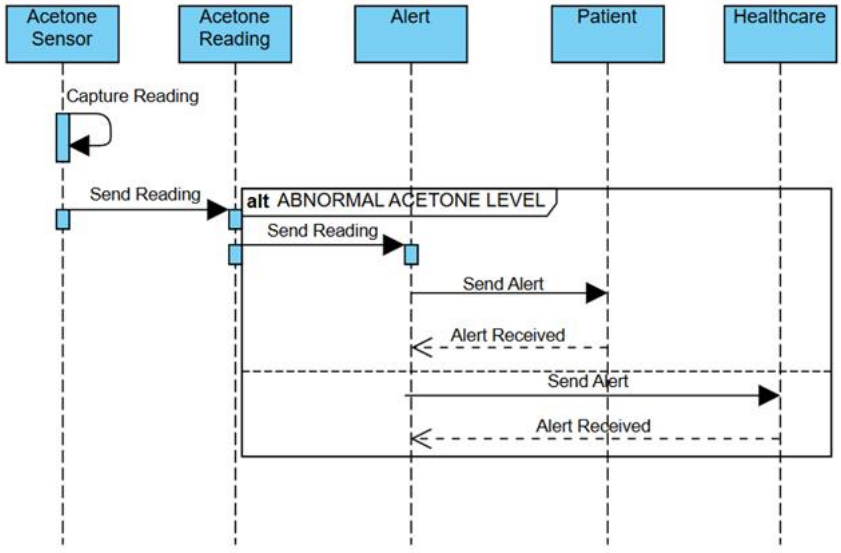
This is an interaction of flow between The Glucose sensor, the glucose reading, Alert, Patient and healthcare provider, this is a sequence diagram of FR1

The sequence diagram shows the entire timeline and flow and the sequence of events

The Sequence starts with the glucose sensor capturing and collecting glucose readings from the diabetic patient it is connected to, the next thing it does once it collects them is turning them to glucose reading that needs to pass a test of threshold of whether they don't breach the pre-set levels, once they are sent as readings, the it's checked if the reading passes the predetermined threshold glucose or not through an internal logic gate that detects the value, if it passes it starts an alternative route, the glucose reading will trigger an alert about the glucose level that is sent to the patient to their mobile application alongside the glucose reading as an alert notification, once it arrives it is validated to the system by the patient receiving the alert and the glucose reading, at the same time an alert was sent to the healthcare provider, also

validated and then successfully received by the healthcare provider.

- **Nour Marai Part in Documentation of The Sequence Model**



The interaction flow between the Acetone Sensor, Acetone Reading, Alert, Patient and Healthcare Provider classes in the healthcare monitoring is shown in the sequence diagram of FR2. The sequence diagram shows the time order of the operations that are performed in the healthcare system to monitor the concentration of acetone and to send warning notifications for the healthcare in case the concentration of acetone is abnormal.

The process of monitoring health care starts from the moment the Acetone Sensor executes Capture Reading operation, continuously capturing the level of acetone in the patient. Once the concentration value of the acetone is acquired, the Acetone Sensor will transmit the acetone concentration reading to the Acetone Reading class via the Send Reading operation.

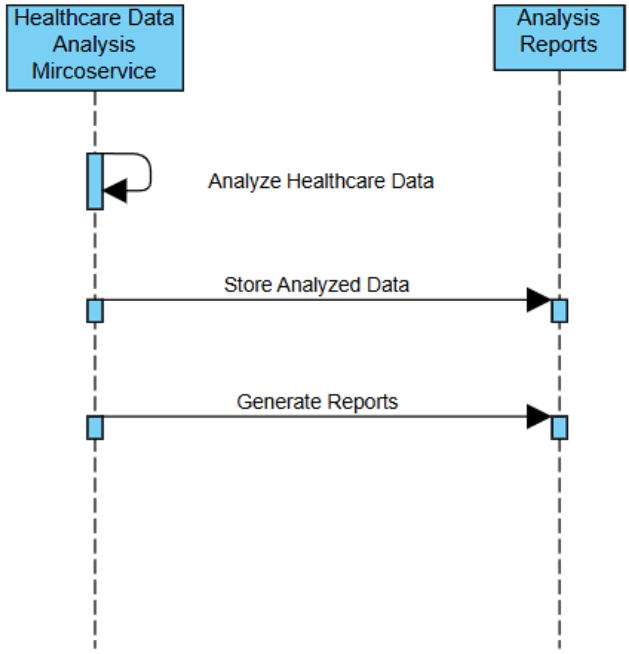
The alternate path, “ABNORMAL ACETONE LEVEL” is then tested on the sequence diagram. If the Acetone concentration is abnormal, the

Acetone Reading class activates the Alert class by sending a Reading message to start the Healthcare Notification process.

The Send Alert operation is then sent by the Alert class to the Patient and Healthcare Provider classes. Both entities then send a confirmative notification of a received healthcare warning by sending an Alert Received reply message.

The sequence diagram illustrates the flow of operations related to real time healthcare monitoring and healthcare alert interaction in FR2 and represents the operations required for continuous monitoring and alert notification of diabetic healthcare in the healthcare monitoring system.

• Youssef Alaa Part in Documentation of The Sequence Model



This Sequence Diagram shows the interaction between the Healthcare Data Analysis Microservice and the Analysis Reports class. The Healthcare Data Analysis Microservice performs a self-call to execute the

action Analyze Healthcare Data. In this step, the microservice analyzes glucose and acetone data using dual-biomarker analytical algorithms to determine the metabolic risk level, which can be Normal, Warning, or Critical.

After the analysis is completed, the microservice sends the analyzed healthcare data to the Analysis Reports class through the Store Analyzed Data message. The stored data includes the risk level, glucose value, acetone value, timestamp, and patient ID.

Then, the microservice sends another message to the Analysis Reports class to Generate Reports. This allows the system to create healthcare reports based on the analyzed data and store them for future monitoring and remote healthcare follow-up.

Reflected Functional Requirements

Analyze healthcare data

Reflected Use Cases

Analyze healthcare data

Store analyzed data

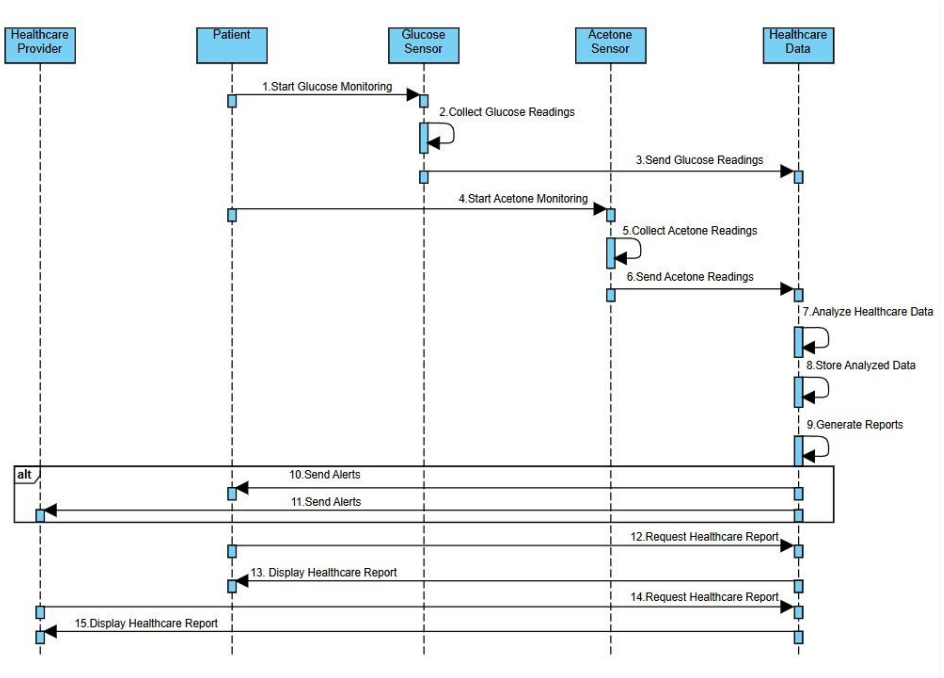
Generates reports

Reflected Classes

Healthcare Data Analysis Microservice

Analysis Reports

5.5.2 Microservices Integration in Sequence diagram



The Microservices Integration Sequence Diagram shows the sequence of interactions between the microservices involved in continuous glucose monitoring, acetone monitoring, healthcare data analysis, healthcare alert generation, and healthcare reporting in the smart healthcare monitoring system. The sequence diagram illustrates how FR1, FR2, and FR3 are combined to provide real-time monitoring and early detection of diabetic ketoacidosis (DKA) for diabetic healthcare.

The glucose monitoring workflow starts when the Patient initiates the glucose monitoring process via the “Start Glucose Monitoring” operation. The Glucose Sensor next executes the “Collect Glucose Readings” operation, which gets the patient's glucose concentration values continuously. Once the glucose results are received, the Glucose Sensor calls the “Send Glucose Readings” in the Healthcare Data microservice to send the healthcare data generated.

Likewise, the Patient starts the monitoring of Acetone with the operation "Start Acetone Monitoring". The Acetone Sensor is then used to execute the operation "Collect Acetone Readings", which is to continuously record the values of the concentrations of acetone by using the wearable acetone skin biosensor embedded in the health monitoring system. The Acetone Sensor then sends the acetone readings to the "Send Acetone Readings" operation of the Healthcare Data microservice.

Once the Healthcare Data microservice receives both the glucose concentration and the acetone concentration, it executes the "Analyze Healthcare Data" operation to analyze the healthcare data and identify the abnormal healthcare patterns linked with the risk of DKA. The analysed healthcare data is then processed and stored using the "Store Analyzed Data" operation. In addition, the "Generate Reports" operation is called on the Healthcare Data microservice to generate Healthcare monitoring reports, derived from the analyzed Healthcare Data information.

The sequence diagram is then tested for the other condition, abnormal healthcare conditions. If the abnormal concentration level of glucose and acetone exceeds the predefined Healthcare standards, the "Send Alerts" operation is called by the Healthcare Data microservice to send out real-time alerts to the Healthcare Provider and Patient.

Furthermore, the Patient and Healthcare Provider can call "Request Healthcare Report" operation on the Healthcare Data microservice to request healthcare monitoring reports. Healthcare reports generated are then presented to both entities via the "Display Healthcare Report" operation to assist in the continuous monitoring of healthcare and medical follow-up activities.

The Microservices Integration Sequence Diagram displays the flow of integration and interaction between the healthcare monitoring

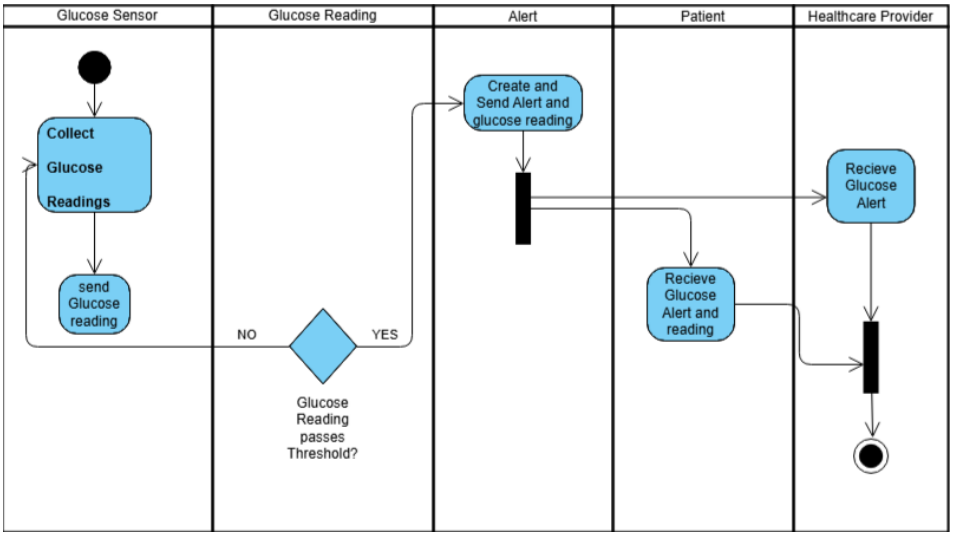
microservices, and how the three functional requirements (FR1 “Monitor Glucose Levels”, FR2 “Monitor Acetone Levels”, and FR3 “Analyze Healthcare Data”) interact with each other to enable intelligent real-time healthcare monitoring in the envisioned smart healthcare monitoring system.

5.6. Activity Diagram

5.6.1. Activity Diagram Model

5.6.2. Documentation of Activity Model

• Ali Nagy Part in Documentation of The Activity Model



Explanation:

This is an activity diagram of the FR1, to show the activities and their workflow, this is a workflow for glucose monitoring, the activity diagram shows the following:

It shows the order of the activities that are carried by the glucose sensor, the glucose reading, the alert itself, the patient, and the healthcare provider

The workflow starts by the glucose sensor collecting the readings of glucose in the blood of the diabetic patient connected to it, it then enters the second phase in the glucose reading lane, a logic gate the answers the question “Glucose Reading passes Threshold” if it doesn’t pass the threshold “NO” of dangerous

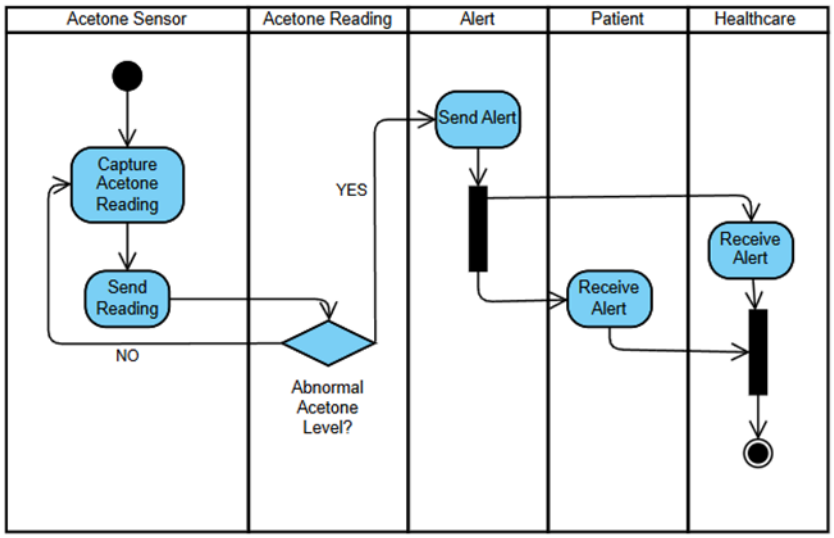
glucose levels in the logic gate, the reading is sent normally and the operation repeats on, if it passes “YES”

The flow moves to the Alert lane a activity where an alert is created and sent through the “Create and Send Alert” action to the healthcare monitoring system

Once the alert is created and sent, The patient receives the alert through their mobile phone application as a notification and the healthcare provider receives the alert on their medical dashboard, both happen through the “Receive glucose Alert” action, which then proceeds to terminate the process and start it all over again,

This is the activity diagram and the workflow from FR1.

- **Nour Marai Part in Documentation of The Activity Model**



The activity diagram of FR2 depicts the workflow of the healthcare monitoring operations related to monitoring the

concentration level of the acetone and issuing a healthcare warning notification in the healthcare monitoring system. The activity diagram shows the order of activities that are carried out by the Acetone Sensor, Acetone Reading, Alert, Patient, and Healthcare Provider components in the healthcare monitoring process.

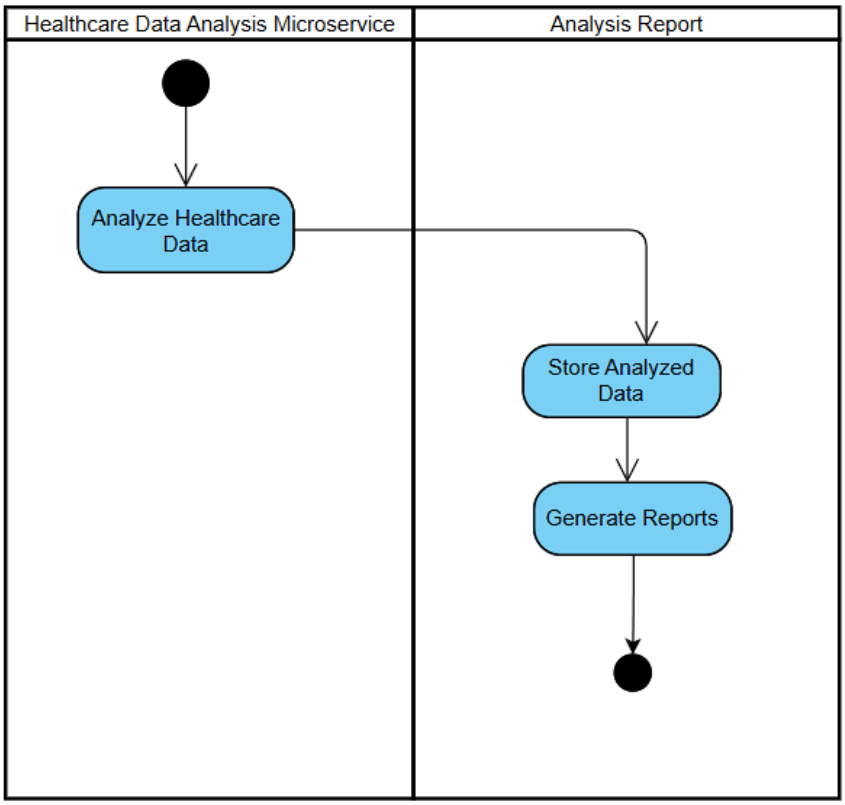
The healthcare monitoring process starts with the activity “Capture Acetone Reading” which is carried out by the Acetone Sensor to continuously read the level of acetone concentration from the patient. Once the value of the Acetone is captured, the Acetone Sensor does the “Send Reading” activity to send the value of the Acetone to healthcare monitoring system.

The workflow then checks the decision condition “Abnormal Acetone Level?”. In the case of the normal level of acetone concentration, the workflow restarts the healthcare monitoring cycle by performing the acetone reading capture process again for the continuous monitoring operations.

If the abnormal levels of acetone concentration are detected, then the workflow continues to the “Send Alert” activity in the Alert component. The “Receive Alert” activities then receive the healthcare warning notification simultaneously to the Patient and Healthcare Provider entity.

The healthcare warning notifications generated by the healthcare monitoring system are finally delivered to the patient and healthcare provider, ending the healthcare monitoring workflow. The activity diagram represents the continuous healthcare monitoring and healthcare alert flow required to enable real time diabetic healthcare monitoring in FR2.

• Youssef Alaa Part in Documentation of The Activity Model



FR3: Analyze Healthcare Data

This Activity Diagram shows the flow of the Analyze Healthcare Data activity performed by the Healthcare Data Analysis Microservice. The black initial node represents the start of the activity. The process starts when the microservice receives glucose and acetone data from FR1 and FR2. Then, the microservice analyzes the healthcare data using dual-biomarker analytical algorithms to determine the metabolic risk level, which can be Normal, Warning, or Critical. After the analysis is completed, the analyzed data is sent to the Analysis Report class to store the results in the cloud database and generate healthcare reports.

Reflected FR (Functional Requirements):	Analyze Healthcare Data
Reflected Use Case:	• Analyze healthcare data • Store analyzed data • Generate reports
Reflected Class:	• Healthcare Data Analysis Microservice • Analysis Report

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Quality Map	
Have we built the right product?	Yes. At this stage, the project demonstrates a complete and integrated smart healthcare monitoring solution capable of supporting real-time diabetic healthcare monitoring, healthcare data analysis, healthcare reporting, and healthcare alert generation. The sequence diagrams, activity diagrams, and microservices integration models confirm that the healthcare monitoring workflows support the intended healthcare objectives and operational scenarios identified throughout the project. The integrated interaction between

	glucose monitoring, acetone monitoring, and healthcare data analysis validates that the proposed system effectively supports early diabetic ketoacidosis (DKA) detection and intelligent healthcare monitoring operations.
Have we built the product right?	Yes. The project models and documentation maintain strong consistency, integration, and traceability across all project phases. The sequence diagrams correctly reflect the operations defined within the class diagrams and use cases, while the activity diagrams accurately represent the healthcare workflow execution and decision-making processes. Furthermore, the traceability analysis confirms that the functional requirements, DFD processes, ERD entities, UML models, healthcare workflows, and system operations are fully interconnected and consistently documented. The microservices integration design also demonstrates that the system architecture supports scalable and modular healthcare monitoring operations aligned with the proposed healthcare solution

6. Validation and Verification
   6.1. The Validation Test
   6.1.1. Forward Traceability
- **Ali Nagy's Part**

Symbol	Meaning
P	Patient
H	Healthcare Provider
G	Glucose Sensor
FR1.1	Monitor Glucose Levels
FR1.2	Collect Glucose Readings
FR1.3	Send Alerts

Actor	User Requirements	Forward Traceability
P	The Patient wears the Glucose biosensor that check their blood for glucose levels and then sends them an alert on their mobile phone when glucose levels are dangerous (Passed the threshold)	• FR1 • FR1.2 • FR1.3
H	The healthcare providers monitors the glucose levels and will get warnings whenever the	• FR1 .1 • FR1.3

	pre-set glucose threshold is passed	
G	The Glucose sensor will constantly and continuously detect glucose levels in the blood and send them to the healthcare monitoring system	• FR1.2

- **Nour Marai Part**

The verification test verifies the correctness of the development of all system models and design artifacts, and that they are traceable to the functional requirements of the healthcare monitoring system. Furthermore, the verification process can be used to get consistency between the use case diagram, DFD, ERD, class diagram, sequence diagram and activity diagram of FR2.

Keys:

Symbol	Meaning
P	Patient
H	Healthcare Provider
A	Acetone Sensor
FR2.1	Monitor Acetone Levels
FR2.2	Collect Acetone Readings
FR2.3	Send Alerts

Actor	User Requirements	Forward Traceability
P	The patient can wear the device that contains acetone biosensors to check the concentration level of acetone and receives warning alerts when the concentration is abnormal.	➢ FR2 ➢ FR2.1 ➢ FR2.2

H	Healthcare provider receives health warnings while patient's acetone is abnormal and remotely monitors patient's acetone.	➢ FR2 ➢ FR2.2
A	The acetone sensor captures and sends acetone concentration readings continuously to the healthcare monitoring system.	➢ FR2.1

- **Youssef Alaa Part**

Symbol	**Meaning**
P	**Patient**
H	**Healthcare Provider**
FR3.1	Analyze Healthcare Data
FR3.2	Store Analyzed Data
FR3.3	Generate Reports

Actor	**User Requirements**	**Forward Traceability**
P	The patient can view analyzed healthcare results, risk level, abnormal trend reports, and monitoring reports through the mobile healthcare application.	● FR3 ● FR3.1 ● FR3.2 ● FR3.3
H	The healthcare provider can remotely view analyzed patient data, metabolic risk reports, and reports when abnormal glucose	● FR3 ● FR3.1 ● FR3.2 ● FR3.3

	and acetone patterns are detected.	

6.1.2. Backward Traceability

- **Ali Nagy's part**

ID	User Requirements	Backward Traceability
FR1.1	Glucose levels are constantly monitored by a wearable Glucose biosensor by the system, the biosensor and system are connected to a healthcare monitoring platform	• G
FR1.2	Monitored glucose levels are captured and sent to the healthcare monitoring platform constantly	• G
FR1.3	An alert is sent by the system when the threshold of glucose is passed	• P • H

- **Nour Marai Part**

ID	User Requirements	Backward Traceability
FR2.1	The system continuously monitors acetone concentration levels using wearable acetone biosensors connected to the healthcare monitoring platform.	➢ A

FR2.2	The acetone biosensor captures and transmits acetone concentration readings continuously.	➢ A
FR2.3	The system sends healthcare warning notifications when abnormal acetone levels are detected.	➢ P ➢ H

- **Youssef Alaa Part**

ID	**User Requirements**	**Backward Traceability**
FR3.1	The system analyzes glucose and acetone readings in real time using healthcare analytical algorithms to determine the patient metabolic risk level.	● P ● H
FR3.2	The system stores the analyzed healthcare data, including risk level, glucose value, acetone value, timestamp, and patient ID, in the cloud healthcare database.	● P ● H
FR3.3	The system generates healthcare monitoring reports when warning or critical abnormal patterns are detected.	● P ● H

7.1. The Verification Test

7.1.1. Traceability of The Use Case Diagram with The Scope

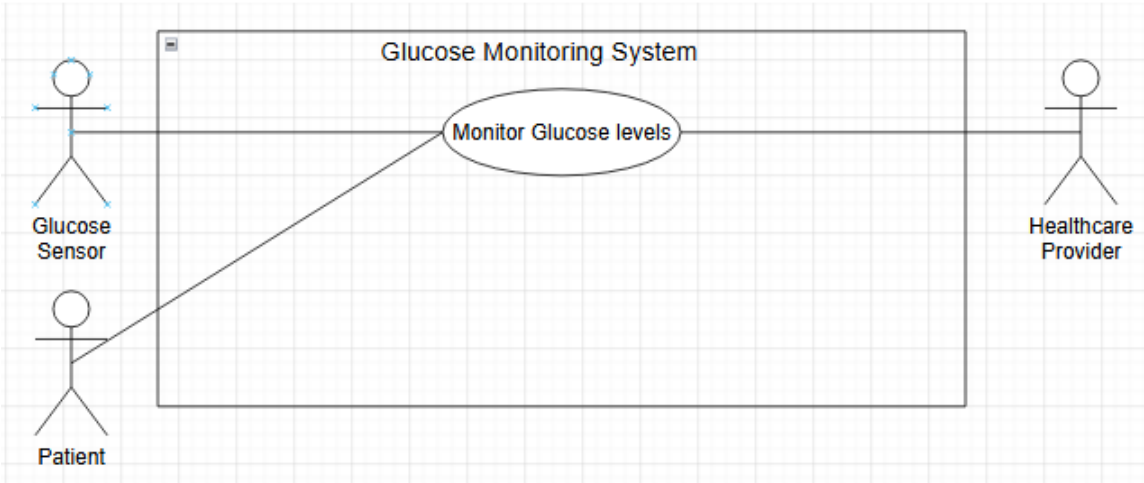
• Ali Nagy Part

FR1: Monitor Glucose levels:

The Healthcare monitoring system is designed to facilitate the the management of diabetes and it's symptoms through a full real-time monitoring by biosensors, one of these biosensors being the glucose sensor, the system allows the patient and their healthcare provider to monitor their own blood glucose rea-time in order to be a head on everything, monitoring glucose levels in the blood, the glucose biosensors sends the collected readings

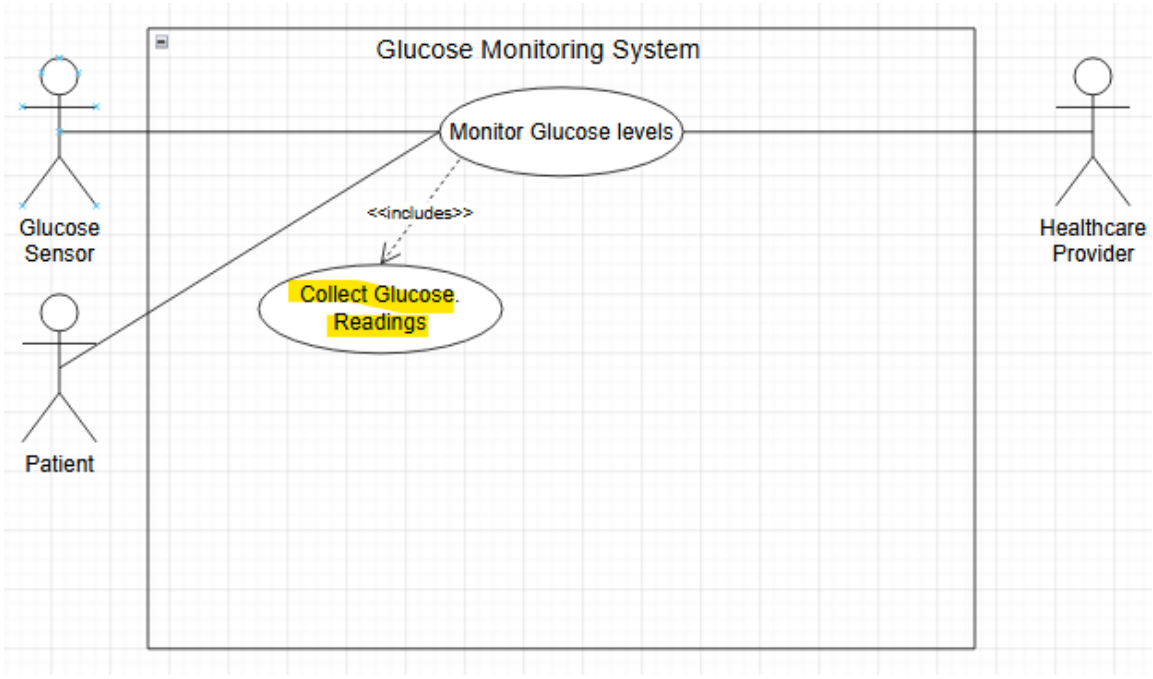
The Use Case Diagram reflects the scope of the system, the actors referring to each parameter in the system, the "Monitor Glucose level" is the main use case and it represents the constant capturing of the glucose readings and testing them against the threshold, the "Collect Glucose Readings" use case sends the readings the to system in order to be accessible, the "Send Alerts" use case represents the notification alerts that reach the healthcare provider and the patient when the glucose reading passes the normal thresholds.

FR.1.1: Monitor Glucose Levels:



The “Monitor Glucose levels” Use case reflects the healthcare monitoring part in the scope, and this is achieved through the present on time monitoring of the concentration levels of glucose in the blood of the patient, this use case facilitates tracking blood sugar for diabetic patients and decreases the risk of accidental overdose insulin intake or forgetting the insulin, this use case automates blood sugar tracking.

FR1.2: Measure Glucose Readings:

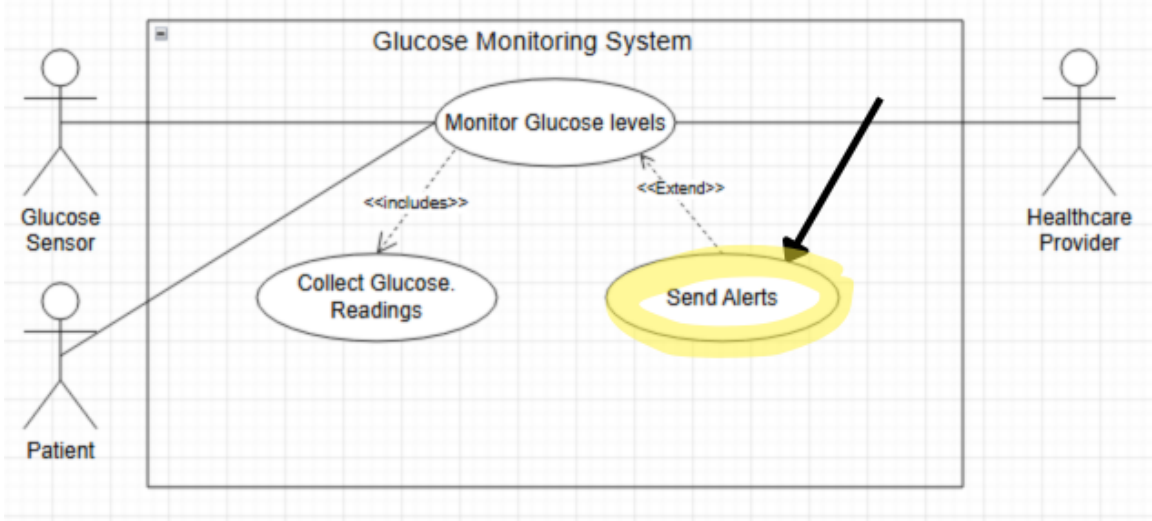


FR1.2: use case “Measure Glucose readings” reflects the scope of healthcare data collection by making the healthcare monitoring system able to receive the glucose readings captured by the sensor consistently, this FR is about the support of automation of the transmission of patient data.

FR.1.3: Send Alerts:

FR1.3 “Send Alerts” tackles the part of the scope that is concerned with the notification, this enables the healthcare monitoring system to send alerts as notifications whenever the

dangerous threshold of glucose have been passed.



- **Nour Marai Part**

FR2: Monitor Acetone Levels

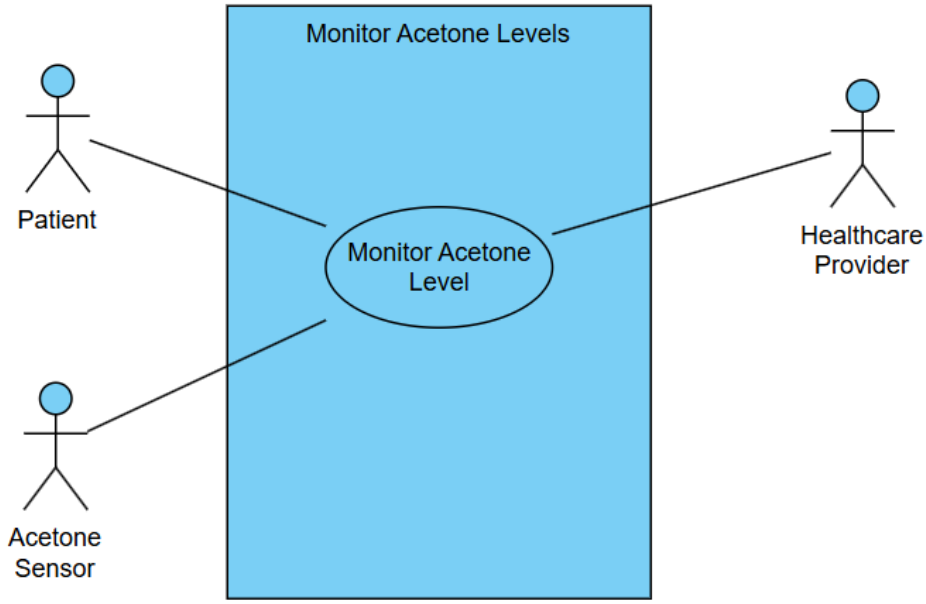
The healthcare monitoring system is designed for continuous real-time monitoring of diabetic patients by using wearable biosensors with healthcare monitoring technologies. This system allows patients and healthcare providers to continuously monitor the concentration of acetone in the system and receive an alert regarding a patient's health condition if concentrations of acetone are outside of normal limits.

This scope is reflected in FR2 “Monitor Acetone Levels” which allows the application of the acetone biosensor for continuous monitoring of acetone concentration and sending the results to the healthcare monitoring platform for monitoring and alert generation.

The Use Case Diagram reflects the healthcare monitoring scope through the interaction between the Patient, Healthcare Provider, and Acetone Sensor actors with the FR2 use cases. The “Monitor Acetone Levels” use case represents the core monitoring functionality, while “Collect Acetone

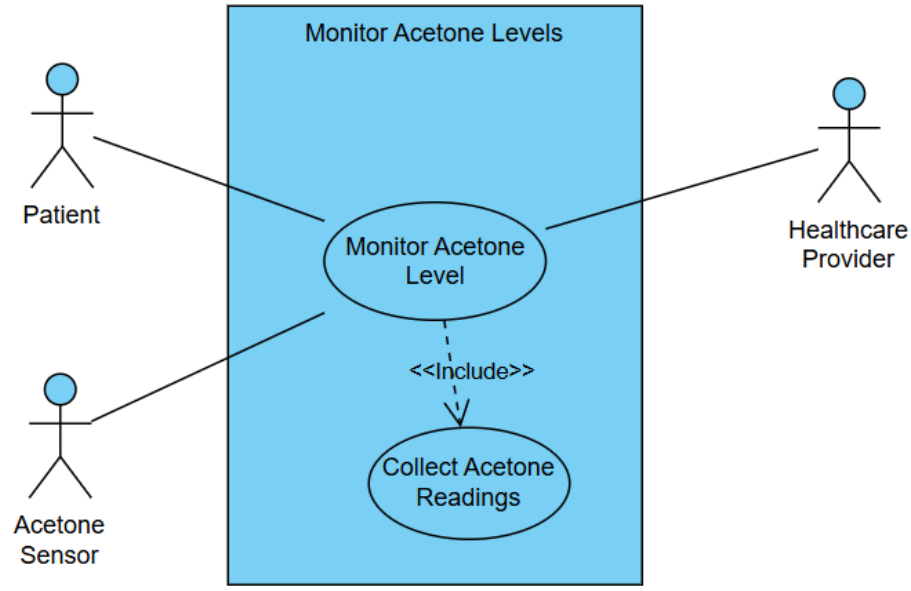
Readings" supports continuous monitoring operations and "Send Alerts" reflects the healthcare notification functionality of the system.

FR2.1 – Monitor Acetone Levels
FR2.1 "Monitor Acetone Levels" reflects the healthcare monitoring scope of the project by enabling continuous real-time monitoring of acetone concentration levels through wearable acetone biosensors integrated with the healthcare monitoring platform. This functional requirement supports automated diabetic healthcare monitoring and continuous patient observation.

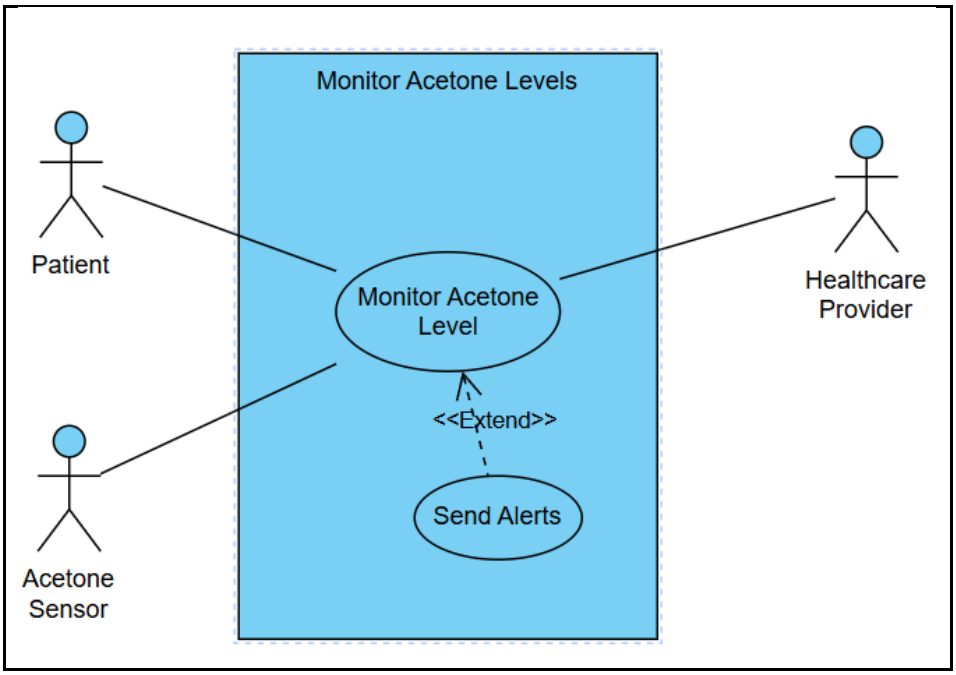


FR2.2 – Collect Acetone Readings
FR2.2 "Collect Acetone Readings" reflects the healthcare data acquisition scope of the project by enabling the healthcare monitoring system to continuously receive acetone concentration readings captured by wearable acetone biosensors. This functional requirement

supports automated healthcare data collection and continuous transmission of patient monitoring data.



FR2.3 – Send Alerts
FR2.3 “Send Alerts” reflects the remote healthcare monitoring and healthcare notification scope of the project by enabling the healthcare monitoring system to send real-time healthcare warning notifications to both patients and healthcare providers whenever abnormal acetone concentration levels are detected.



• Youssef Alaa Part

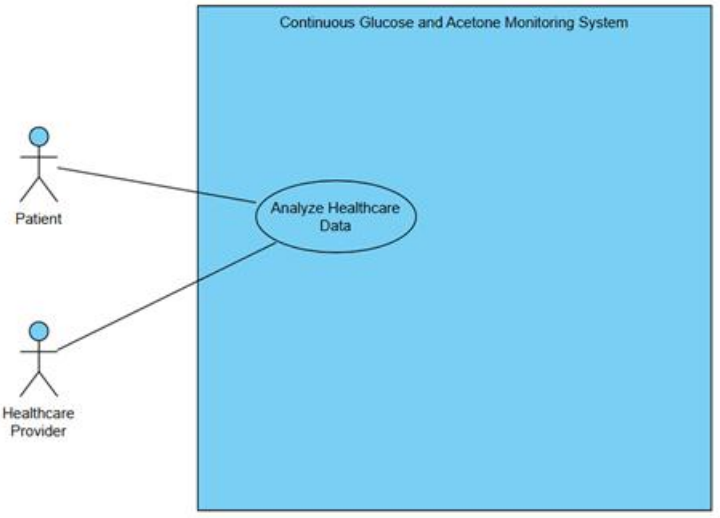
FR3: Analyze Healthcare Data

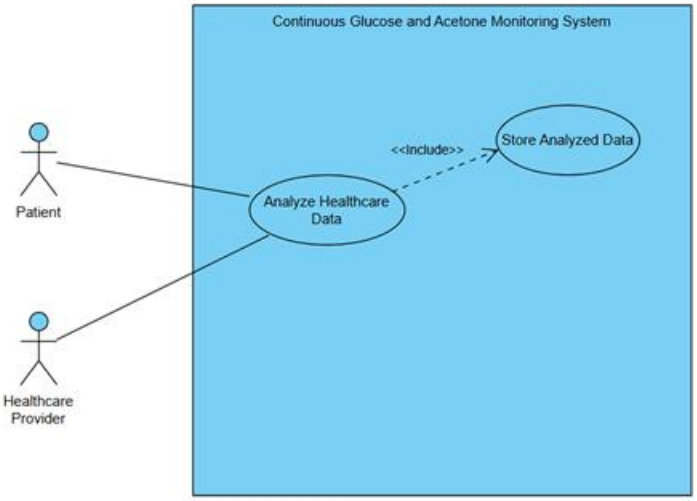
The healthcare monitoring system aims to facilitate intelligent real-time analysis of the diabetic patient data, as glucose and acetone readings from both FR1 and FR2 are processed. This system enables the patient and healthcare professionals to receive analysis of healthcare results, metabolic risk classifications, monitoring reporting on abnormal healthcare patterns.

This scope is manifested in FR3 “Analyze Healthcare Data”, which allows the healthcare platform to use analytical algorithms to process glucose and acetone readings, detect abnormal metabolic trends, classify the patient's risk level as Normal, Warning or Critical, and provide reports for medical follow-up.

The Use Case Diagram shows the scope of healthcare data analysis represented by the interaction between the Use Cases of the healthcare provider (FR3) and the Patient. The use case “Analyze Healthcare Data”

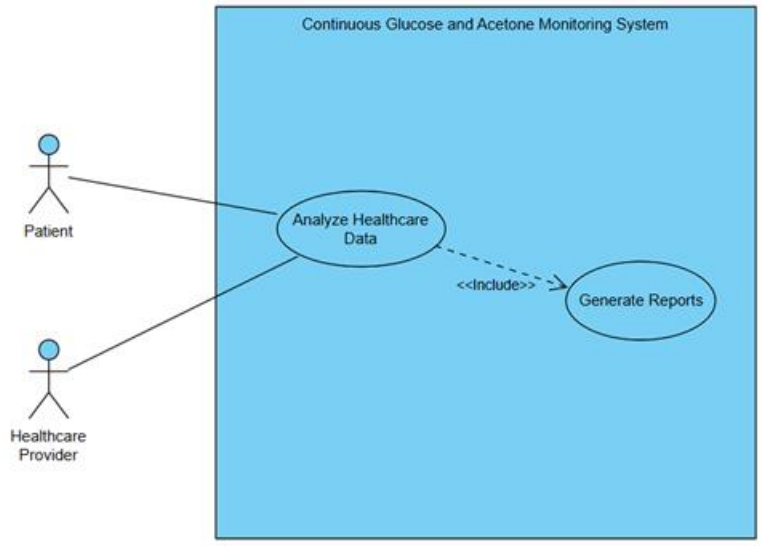
is the main analysis function and the use case “Store Analyzed Data” enables the saving of the analysis results in the healthcare cloud database. Generate Reports” and “Display Reports” refer to the healthcare reporting capabilities and to display the analyzed information to the patient via the mobile application and to the healthcare provider via the web dashboard, respectively.

FR3.1 – Analyze Healthcare Data
FR3.1 “Analyze Healthcare Data” The scope of the project is in healthcare data analysis, which is demonstrated by analyzing glucose and acetone data in real time. The system detects abnormal health patterns based on the readings received from FR1 and FR2 and identifies abnormal healthcare pattern. This functional requirement enables a better understanding of the health condition of the patient between the patient and healthcare providers and supports intelligent healthcare monitoring. The analyzed results are presented via the mobile application and healthcare provider dashboard and reports are generated based on abnormal reading or health pattern.   

FR3.2 – Store Analyzed Data
FR3.2 “Store Analyzed Data” The project scope of healthcare data storage is shown in FR3.2 “Store Analyzed Data” where the system can store the analysed glucose and acetone data in the cloud-based health care database. This information recorded contains the patient's ID, glucose level, acetone level, time and risk level. This functional requirement enables monitoring of healthcare remotely by maintaining the analysis results for future inspection. It also enables healthcare providers to keep an eye on the patient's health history and monitor any changes in the patient's condition over time. 

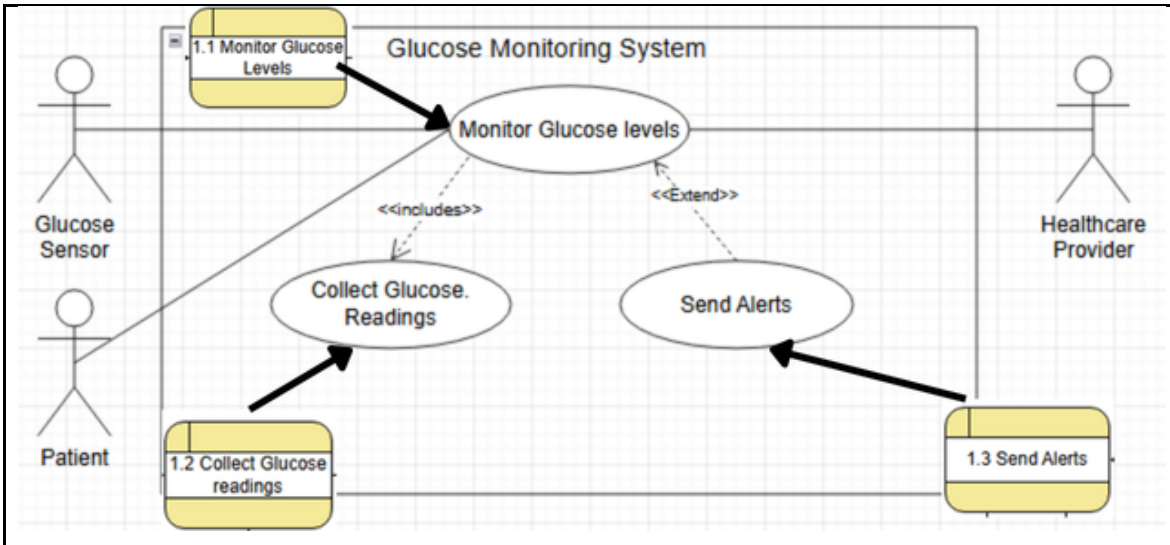
FR3.3 Generate Reports
FR3.3 “Generate Reports” is a feature of the project that is relevant to the healthcare reporting aspect of the project, because as the abnormal pattern of glucose and acetone is detected by the system, it generates a monitoring report. This function enables patients and health care providers to discuss potential health risks. This is used to show monitoring reports to the patient via mobile app and to the healthcare provider via web dashboard, which enables remote healthcare monitoring. It also supports healthcare providers to

follow up on the patient's condition if necessary



7.1.2. Traceability of the 0-level DFD with The Use Case

- **Ali Nagy Part**



FR1.1: Monitor Glucose level: This "1.Monitor Glucose Levels" process in the data flow diagram represents the "Monitor Glucose Levels" Use Case, Which is about the real-time

monitoring of glucose values in the blood of the diabetic patient attached to the sensor, the sensor is detecting the levels of Glucose

FR1.2: Collect Glucose readings: The "Collect Glucose Readings" process in the dataflow diagram represents the use case "Collect Glucose Readings", This process represents the continuous collection of the Glucose readings and their transmission by the wearable Glucose Biosensor in order to send them to the healthcare monitoring platform for operations

FR1.3: Send Alerts: The "Send Alerts" process in the dataflow diagram represents the use case "Send Alerts" in the use case model, this process represents the notification and alerting ability of the healthcare monitoring system whenever dangerous levels of glucose concentration in the blood are detected, the alerts are sent real-time to the healthcare provider through their health dashboard as a notification and to the healthcare patient through a notification on their mobile phone.

- **Nour Marai Part**

FR2.1 – Monitor Acetone Levels

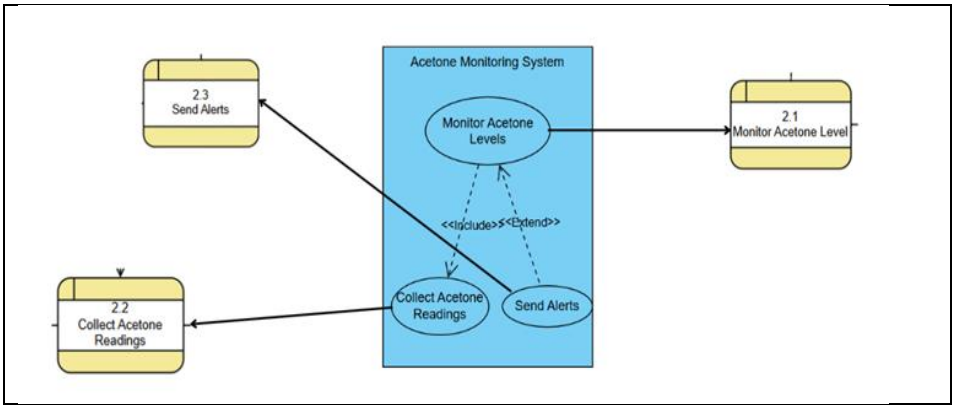
The use case "Monitor Acetone Levels" is reflected in Process 2.1 "Monitor Acetone Levels" in the 0-Level DFD. This process represents the continuous monitoring operation performed by the healthcare monitoring system through wearable acetone biosensors to support real-time diabetic healthcare monitoring.

FR2.2 – Collect Acetone Readings

The use case "Collect Acetone Readings" is reflected in Process 2.2 "Collect Acetone Readings" in the 0-Level DFD. This process represents the continuous collection and transmission of acetone concentration readings captured from wearable acetone biosensors to the healthcare monitoring platform for further monitoring operations.

FR2.3 – Send Alerts

The use case "Send Alerts" is reflected in Process 2.3 "Send Alerts" in the 0-Level DFD. This process represents the healthcare notification functionality of the healthcare monitoring system, where real-time healthcare warning notifications are sent to both patients and healthcare providers whenever abnormal acetone concentration levels are detected.



•Youssef Alaa Part

FR3.1 – Analyze Healthcare Data

The use case "Analyze Healthcare Data" is represented in Process 3.0 "Analyze Healthcare Data" in the 0-Level DFD. This process represents the healthcare data analysis function of the monitoring system. In this process, glucose and acetone values are analyzed by the Healthcare Data Analysis Microservice using analytical algorithms to determine the patient's metabolic risk level, such as Normal, Warning, or Critical.

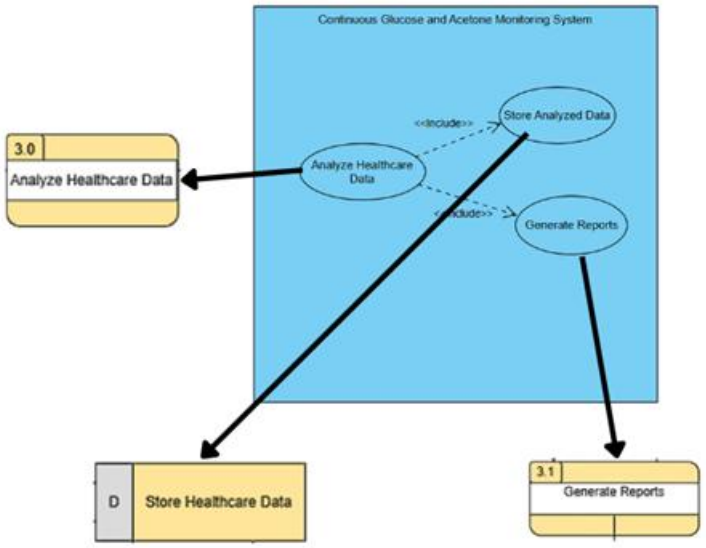
FR3.2 – Store Analyzed Data

The use case "Store Analyzed Data" is represented in the 0-Level DFD through the data flow from Process 3.0 "Analyze Healthcare Data" to the Data Store "Store Healthcare Data." This data store is responsible for saving the analyzed healthcare data after processing. The stored data includes glucose value, acetone value, risk level, timestamp, and patient ID for future review, monitoring, and healthcare follow-up.

FR3.3 – Generate Reports

The use case "Generate Reports" is represented in Process 3.1 "Generate Reports" in the 0-Level DFD. This process represents the healthcare reporting function of the monitoring system. It generates monitoring reports based on the analyzed glucose and acetone readings. These reports can be accessed by the patient through the

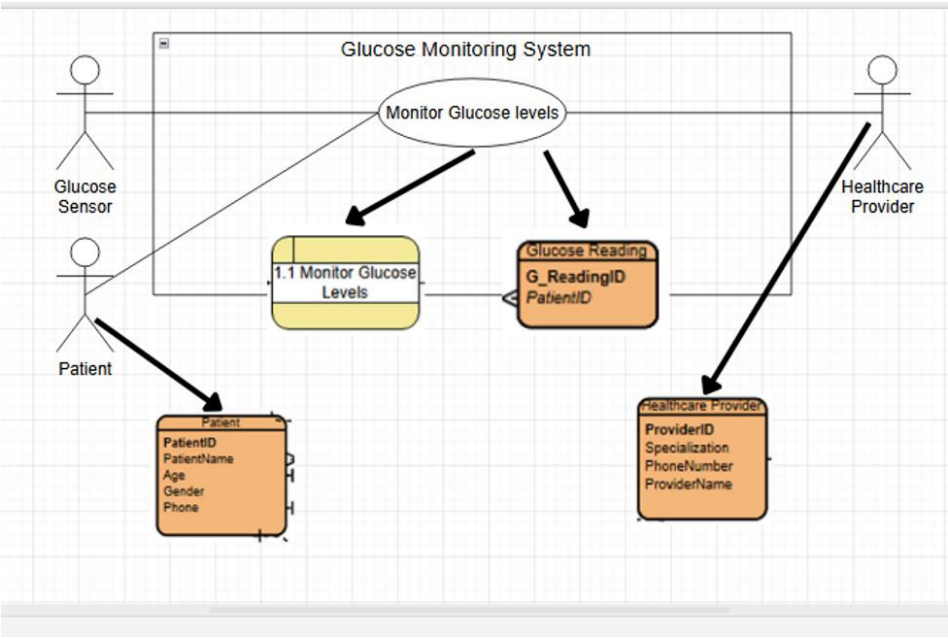
mobile application and by the healthcare provider through the web dashboard.



7.1.3. Traceability of the 0-level DFD and Use Case with the ERD

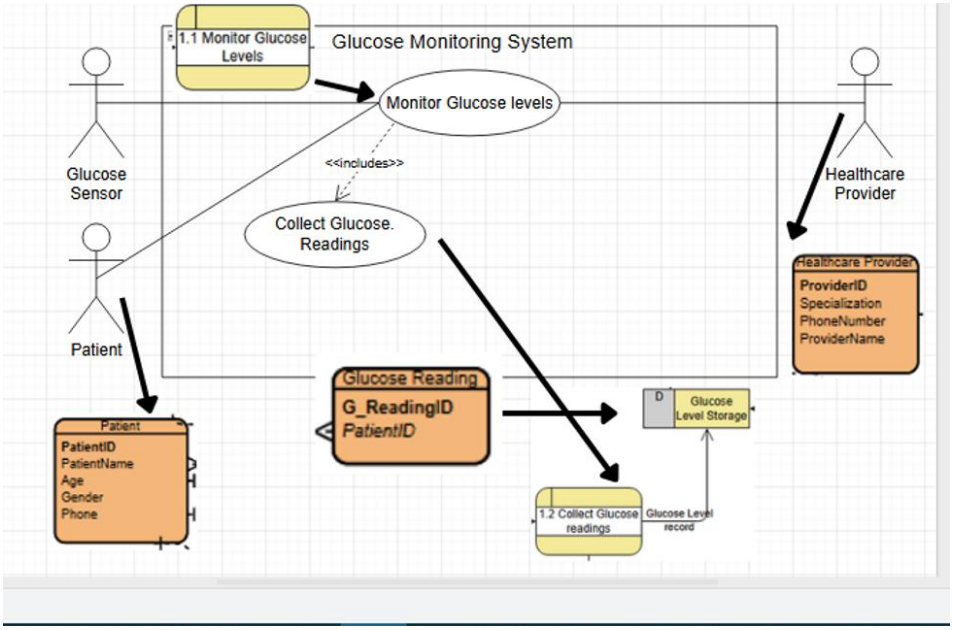
- **Ali Nagy Part**

FR1.1: Monitor Glucose Level



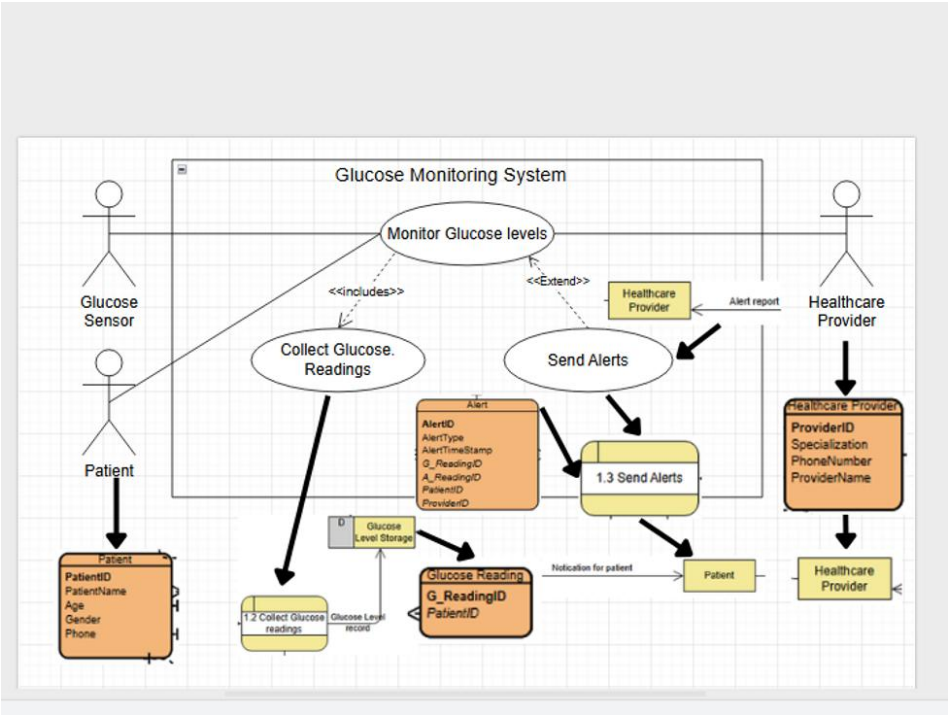
The Monitor Glucose levels is represented in the Use Case Diagram by the “Monitor Glucose levels” Use Case and in the “Monitor Glucose levels” in the activity in the dataflow diagram, both are represented through the Glucose Reading entity in the ERD, with the patient Entity representing the Patient actor in the Use Case Model and with the “Healthcare Provider” actor in the Use Case diagram being represented by the Patient Provider entity, FR1.1 “Monitoring Glucose system” is about monitoring the levels of Glucose in the blood of the patient by the glucose sensor.

FR1.2: Collect Glucose Readings:



"Collect Glucose readings" Use Case in the Use Case Model represents "Collect Glucose readings" activity in the 0 level Dataflow Diagram, this process represents the all-time glucose readings captured by the wearable glucose biosensor and transmission it to the healthcare monitoring platform, this functionality is also reflected in the ERD though the "Glucose Reading" entity, the entity that represents the Glucose values that are gathered and then stored in the local storage shown in the DFD as "Glucose level storage", storing it in the healthcare monitoring system as a backup.

FR1.3: Send Alerts:



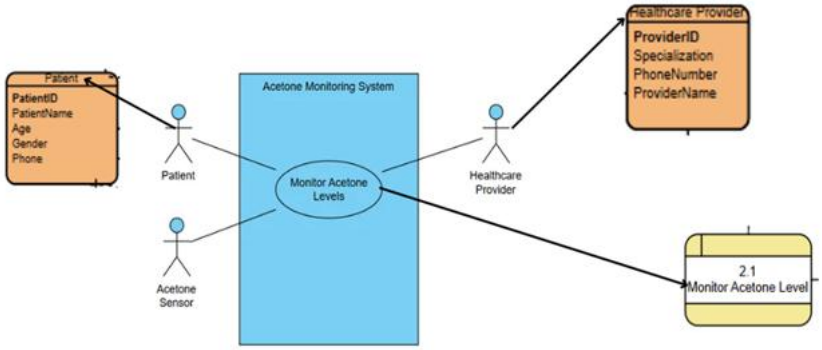
In FR1.3 “Send Alerts” it is reflected by the “Send Alerts” Use Case, which also re[resents the “Send Alerts” activity in the dataflow diagram, when Glucose readings pass the danger threshold or reaches abnormal levels, an alert is sent to the diabetic patient in the form of a notification and to the healthcare provider on their dashboard as a notification, this functionality is represented the ERD through the “Alert Entity” , specifically the “Glucose”

- **Nour Marai Part**

FR2.1 – Monitor Acetone Levels

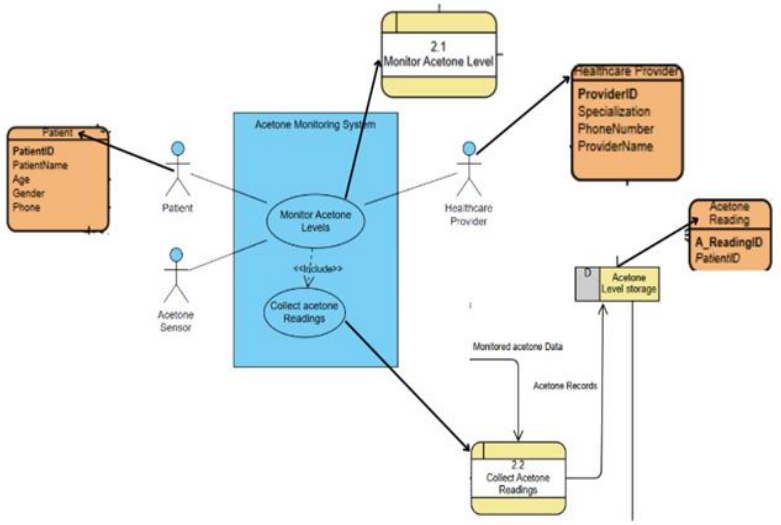
The use case “Monitor Acetone Levels” is reflected in Process 2.1 “Monitor Acetone Levels” in the 0-Level DFD. This process represents the continuous monitoring operation performed by the healthcare monitoring system through wearable acetone biosensors to support real-time diabetic healthcare monitoring. Moreover, this functionality is reflected in the ERD through the Acetone Reading

entity, which stores the acetone concentration readings generated during the healthcare monitoring process.



FR2.2 – Collect Acetone Readings

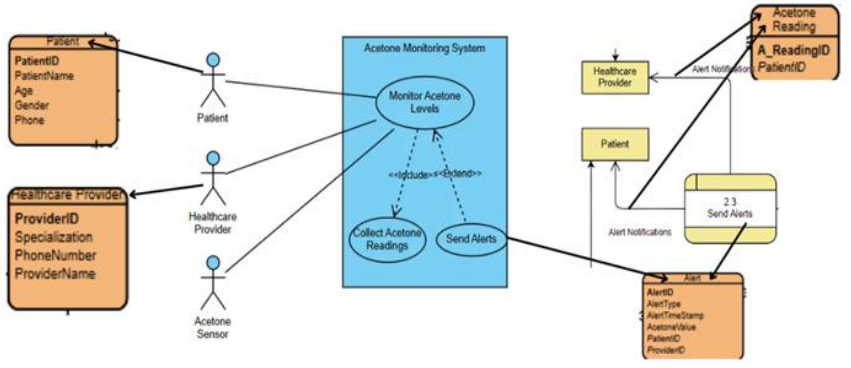
The use case “Collect Acetone Readings” is reflected in Process 2.2 “Collect Acetone Readings” in the 0-Level DFD. This process represents the continuous collection and transmission of acetone concentration readings captured from wearable acetone biosensors to the healthcare monitoring platform for further monitoring operations. Moreover, this functionality is reflected in the ERD through the Acetone Reading entity, which represents the collected healthcare monitoring data stored within the healthcare monitoring system.



FR2.3 – Send Alerts

The use case “Send Alerts” is reflected in Process 2.3 “Send Alerts” in the 0-Level DFD. This process represents the healthcare notification functionality of the healthcare monitoring system, where real-time healthcare warning notifications are sent to both patients and healthcare providers whenever abnormal acetone concentration levels are detected.

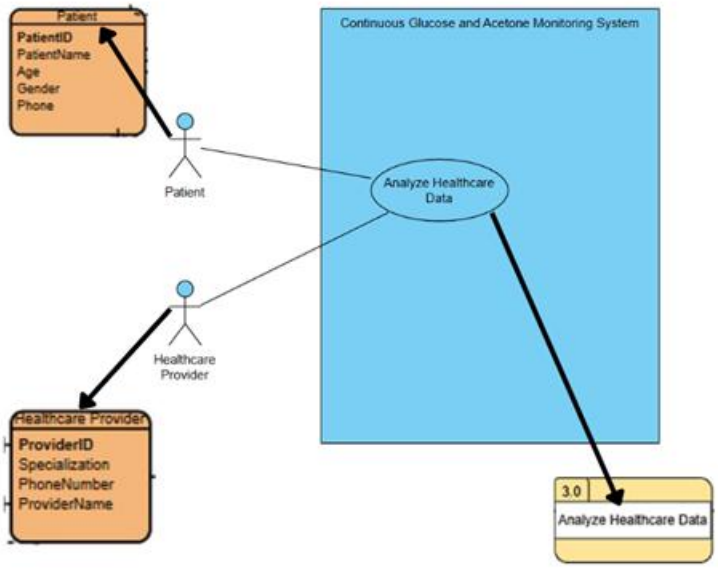
Moreover, this functionality is reflected in the ERD through the Alert, Patient, and Healthcare Provider entities, which represent the healthcare warning notifications and the related healthcare monitoring users within the healthcare monitoring system.



•Youssef Alaa Part

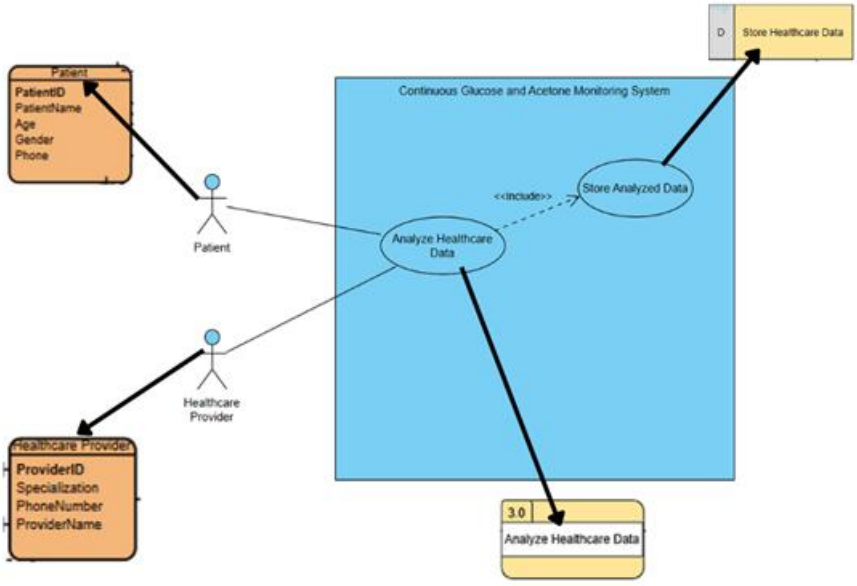
FR3.1 – Analyze Healthcare Data

The use case "Analyze Healthcare Data" is reflected in Process 3.0 "Analyze Healthcare Data" in the 0-Level DFD. This is an operation that is executed by the healthcare monitoring system to analyze the healthcare data during real-time monitoring of health of a diabetic patient, using glucose and acetone readings. Furthermore, this functionality is implemented in ERD as the Healthcare Data entity contains the glucose value of the patient, acetone value, timestamp and risk level derived from the analysis of healthcare data.



FR3.2 – Store Analyzed Data

0-Level DFD contains a Data store called "Store Healthcare Data" which represents the use case "Store Analyzed Data". A data store that contains processed healthcare data that has been analyzed following glucose and acetone readings obtained from the healthcare monitoring system. Furthermore, the ERD depicts this functionality by the entity Healthcare Data, which corresponds to the data of healthcare monitoring that is being analysed by the healthcare monitoring system.

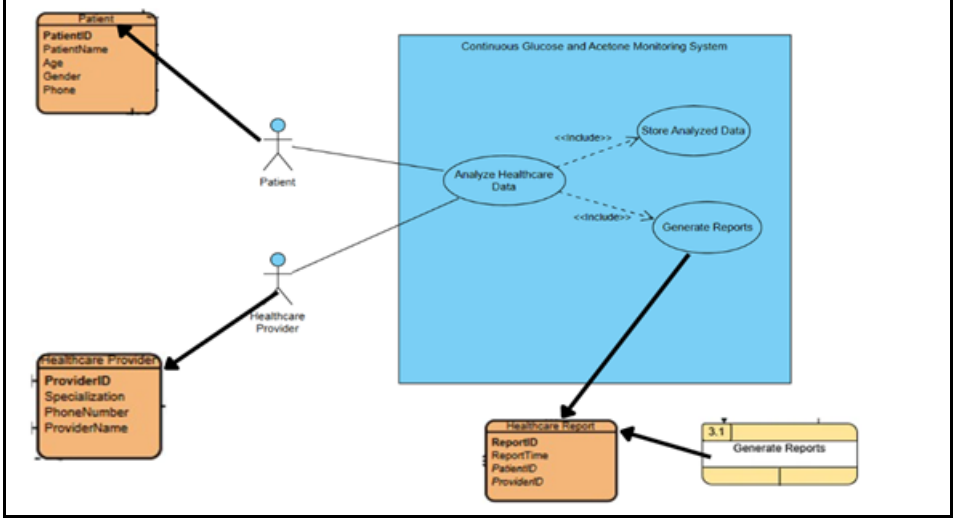


FR3.3 – Generate Reports

The use case "Generate Reports" is reflected in Process 3.1 "Generate Reports" in the 0-Level DFD. This is the healthcare reporting feature of the healthcare monitoring system which generates glucose and acetone monitoring reports after the analysis and provides them to patient and healthcare provider.

Furthermore, this functionality is manifested in the ERD as the Healthcare Report entity, the Patient entity and the Healthcare Provider entity, which correspond to the healthcare monitoring reports generated

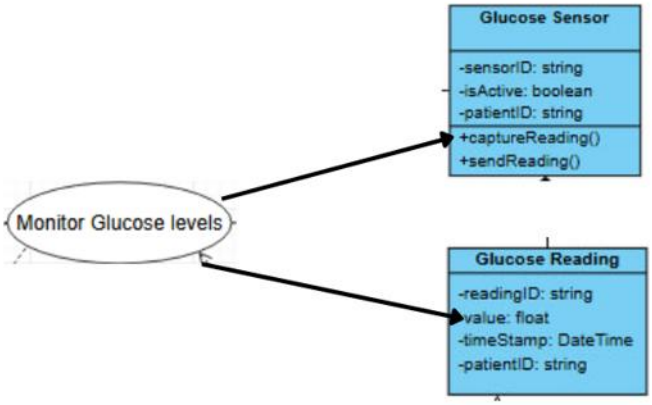
and the users of the healthcare monitoring system.



7.1.4. Traceability of the Class Diagram with the Use Case Diagram

- **Ali Nagy Part**

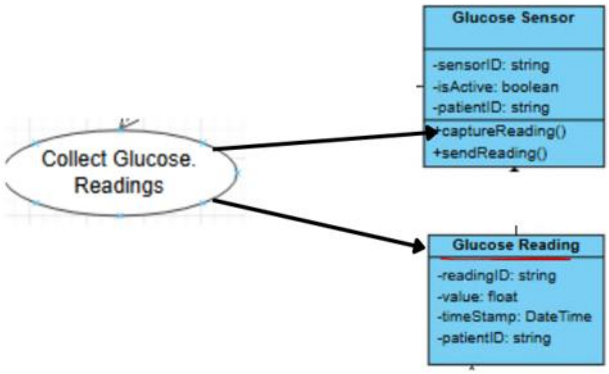
FR1.1: Monitor Glucose Levels:



FR1.1: Monitor Glucose levels is represented by the Use Case “Monitor Glucose Levels inside the Use Case diagram, it is represented in the class diagram as the “CaptureReading()” Operation inside the “Glucose sensor” Class, and the class of “Glucose Reading”, the glucose reading as shown in the class has information about the patient and it’s date of capture.

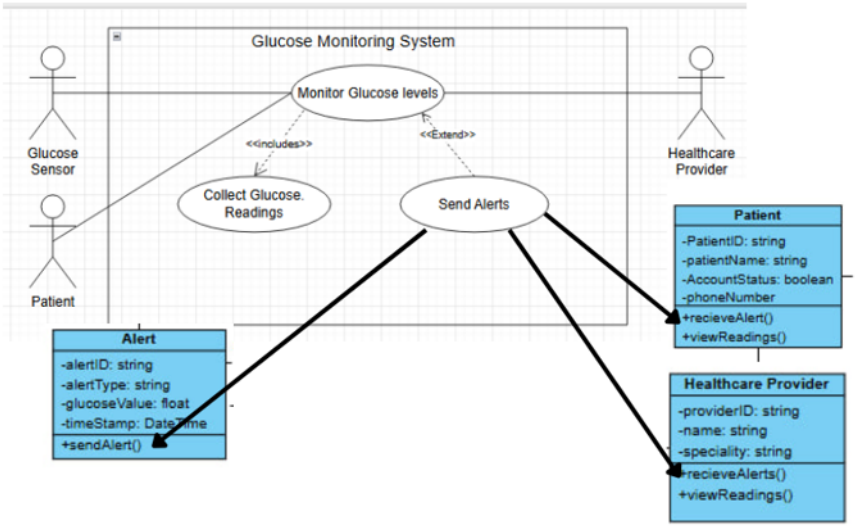
the whole process represents the Glucose sensor monitoring the levels of the Glucose inside the Diabetic Patient’s blood.

FR1.2: Collect Glucose Readings



The Collect glucose readings is represented by the use case “Collect Glucose readings”, glucose readings after being collected and captured by the sensor are sent to the healthcare monitoring system, in the class diagram the operation is defined in the “Glucose Sensor” class Capturing the readings through the “captureReadings()” Operation, the readings captured and stored in the healthcare monitoring system are noted by the class “Glucose Readings”

FR1.3: Send Alerts:



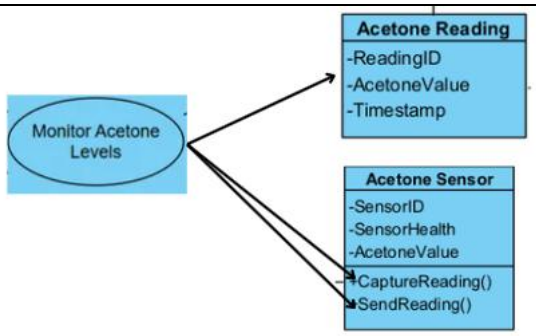
The Send Alert FR1.3 is represented by the "Send Alerts" Use Case, which is mirrored in the Class Diagram in the "Alert" Class.

When Glucose levels break the pre-defined threshold, an alert is sent through the operation "sendAlert()", which then the alert is sent to the Patient and the healthcare proivder and received which is presented in the class diagram as the "receiveAlert()" operation in the "Patient" class and the "Healthcare Provider" class.

- **Nour Marai Part**

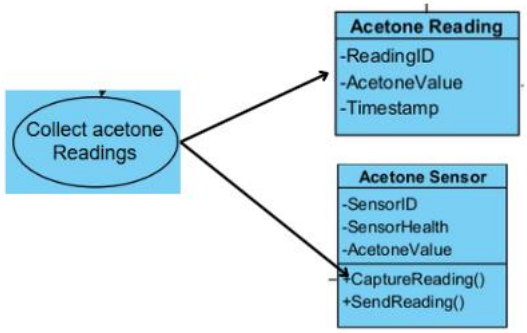
  - The use case "Monitor Acetone Levels" is correlated with the Acetone Sensor and Acetone Reading classes in the class diagram.

    The Acetone Sensor class is responsible for capturing and transmitting acetone concentration readings continuously, while the Acetone Reading class represents the monitoring readings generated during the healthcare monitoring process.



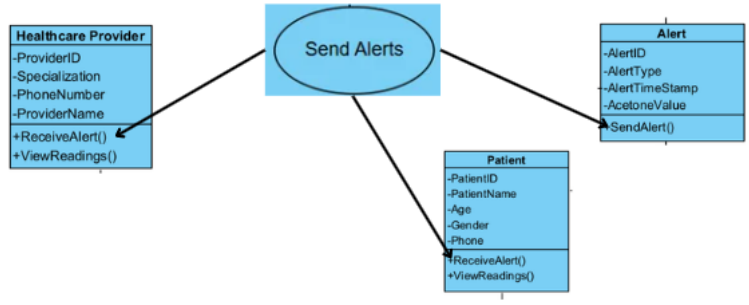
- The use case "Monitor Acetone Levels" is correlated with the Acetone Sensor and Acetone Reading classes in the class diagram.

  The Acetone Sensor class is responsible for capturing and transmitting acetone concentration readings continuously, while the Acetone Reading class represents the monitoring readings generated during the healthcare monitoring process.



- The use case "Collect Acetone Readings" is correlated with the Acetone Sensor and Acetone Reading classes in the class diagram.

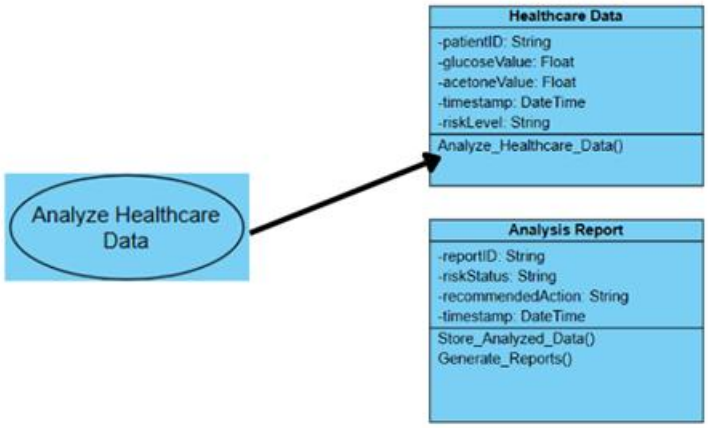
  The Acetone Sensor class performs the reading collection and transmission operations, while the Acetone Reading class stores the collected acetone concentration data used within the healthcare monitoring system.



• Youssef Alaa Part

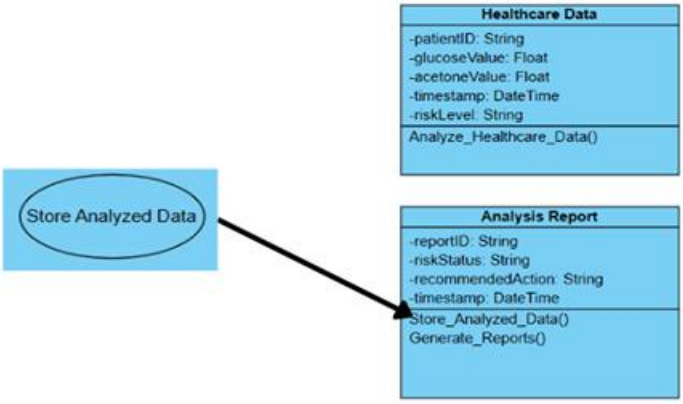
➢ The use case **"Analyze Healthcare Data"** is correlated with the **Healthcare Data** class in the class diagram.

The **Healthcare Data** class is responsible for analyzing the glucose and acetone readings received from the monitoring system. It contains the patient ID, glucose value, acetone value, timestamp, and risk level, which are used to determine the patient's metabolic risk level during the healthcare data analysis process.



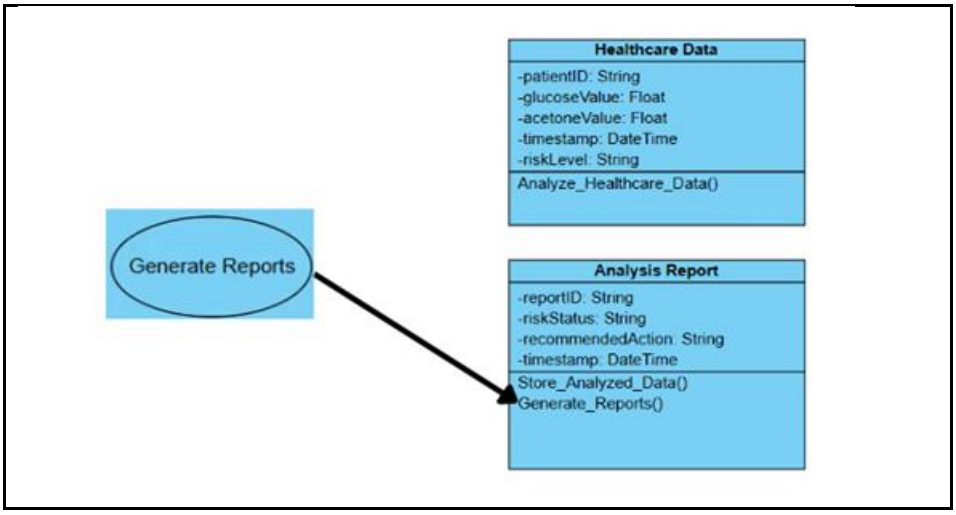
➢ The use case **"Store Analyzed Data"** is correlated with the **Analysis Report** class in the class diagram.

- The **Analysis Report** class is responsible for storing the analyzed healthcare data after the glucose and acetone readings are processed. It contains the report ID, risk status, recommended action, and timestamp, while the **Store_Analyzed_Data()** operation represents the storage function used to save the analysis results for future review and monitoring.



- The use case **"Generate Reports"** is correlated with the **Analysis Report** class in the class diagram.

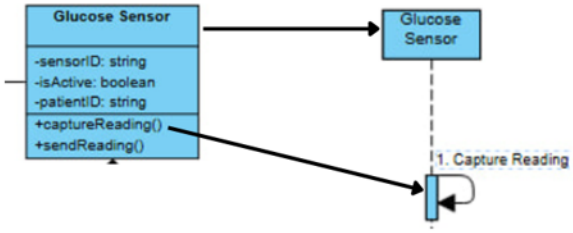
The **Analysis Report** class is responsible for generating healthcare monitoring reports based on the analyzed glucose and acetone readings. It contains the report ID, risk status, recommended action, and timestamp, while the **Generate_Reports()** operation represents the reporting function used to produce reports for future review and healthcare monitoring.



7.1.5. Traceability of the Sequence Diagram with the Class Diagram and the Use Case

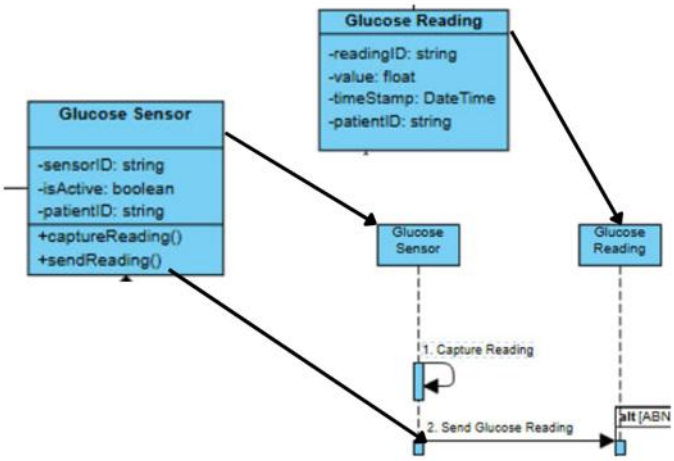
- **Ali NagyPart**

1: Capture reading:



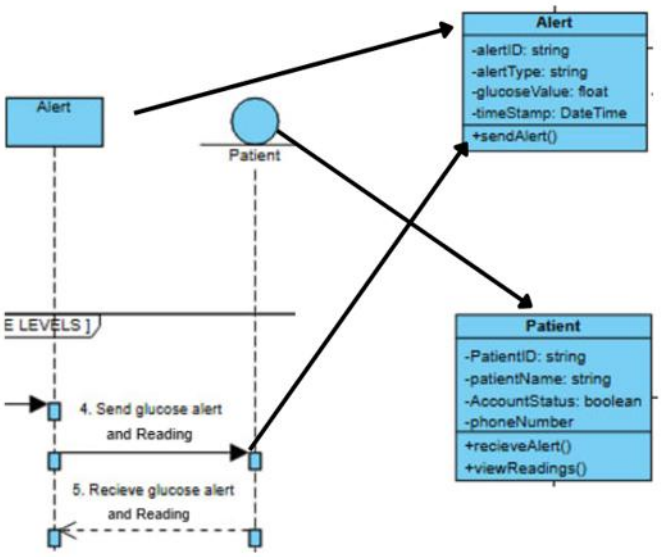
In the figure above, the Glucose sensor in the class diagram is mirrored by the Glucose sensor in the sequence diagram
The first operation “Capture Reading” in the sequence is mirrored by the “captureReadin()” in the Glucose sensor class in the class diagram.

2: Send Glucose Readings:



This sequence here “Send Glucose Reading” is represented by the operation in the Glucose sensor class “sendReading()”
the glucose sensor sends the glucose readings to the healthcare monitoring system.

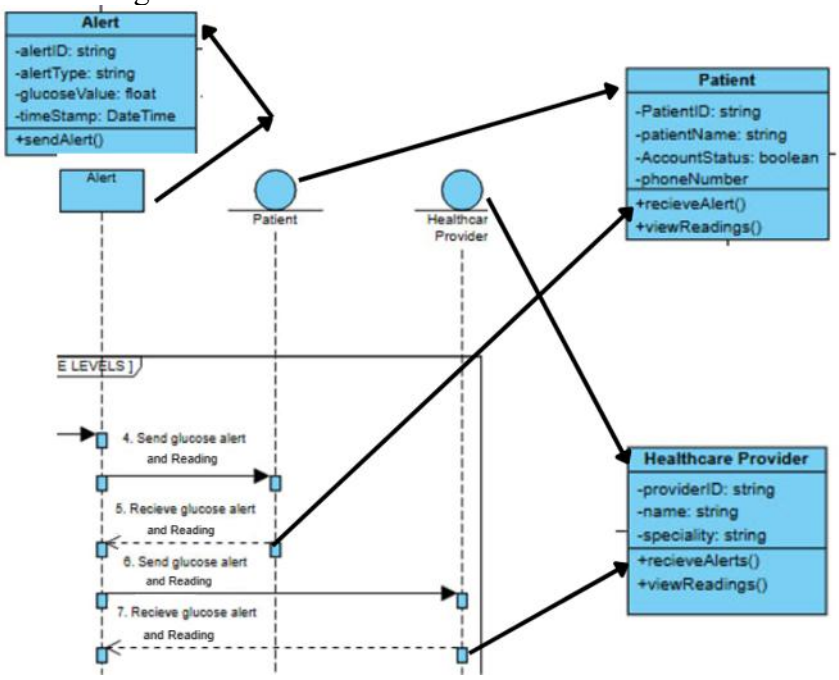
3: Send Glucose Alert



The “Alert” in the sequence diagram is represented by the “Alert” class in the class diagram, the patient is represented by the patient, healthcare provider by healthcare provider, etc Sending the glucose alert in the sequence “Send glucose alert” is mirrored by the class diagram’s operation “sendAlert()”.

After Glucose reaches critical level, an alert is sent by the healthcare monitoring system, which is then send to the healthcare provider and the patient by application.

4. Receive glucose Alert:

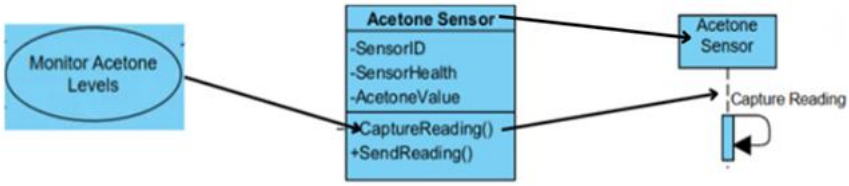


The “Receive Glucose Alert” message in the sequence diagram is represented by the “recieveAlerts()” operation in the “Patient” Class and the “Healthcare Provider” Class

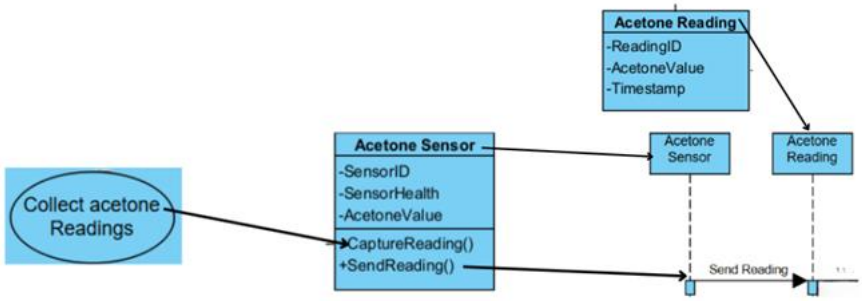
Once The alert is sent, it is validated and received by the patient and the healthcare provider, triggering the notification

- **Nour Marai Part**

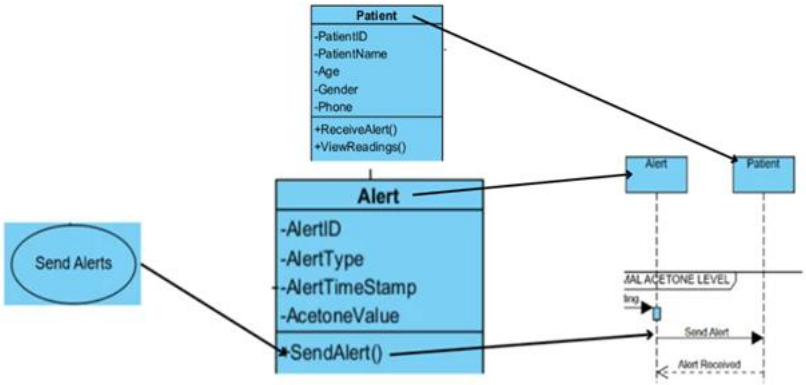
CaptureReading()
Represents the monitoring operation performed by the Acetone Sensor class to capture acetone concentration levels.



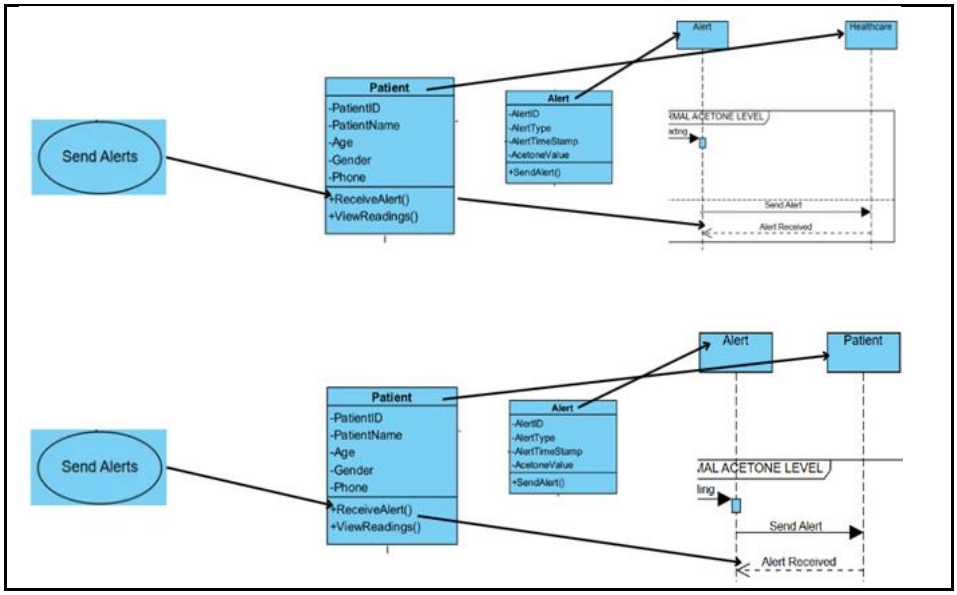
SendReading()
Represents the transmission operation performed by the Acetone Sensor class to send acetone readings to the healthcare monitoring system.



SendAlert()
Represents the healthcare notification operation performed by the Alert class whenever abnormal acetone levels are detected.



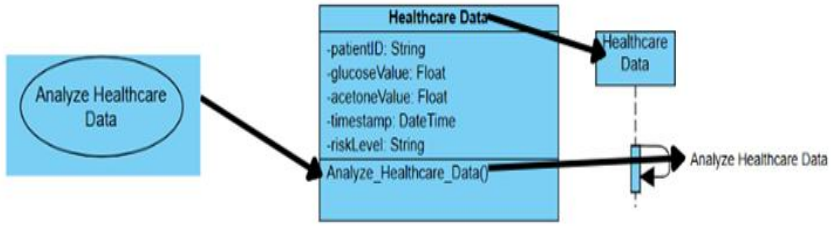
ReceiveAlert()
Represents the healthcare warning notification received by the Patient and Healthcare Provider classes.



• Youssef Alaa Part

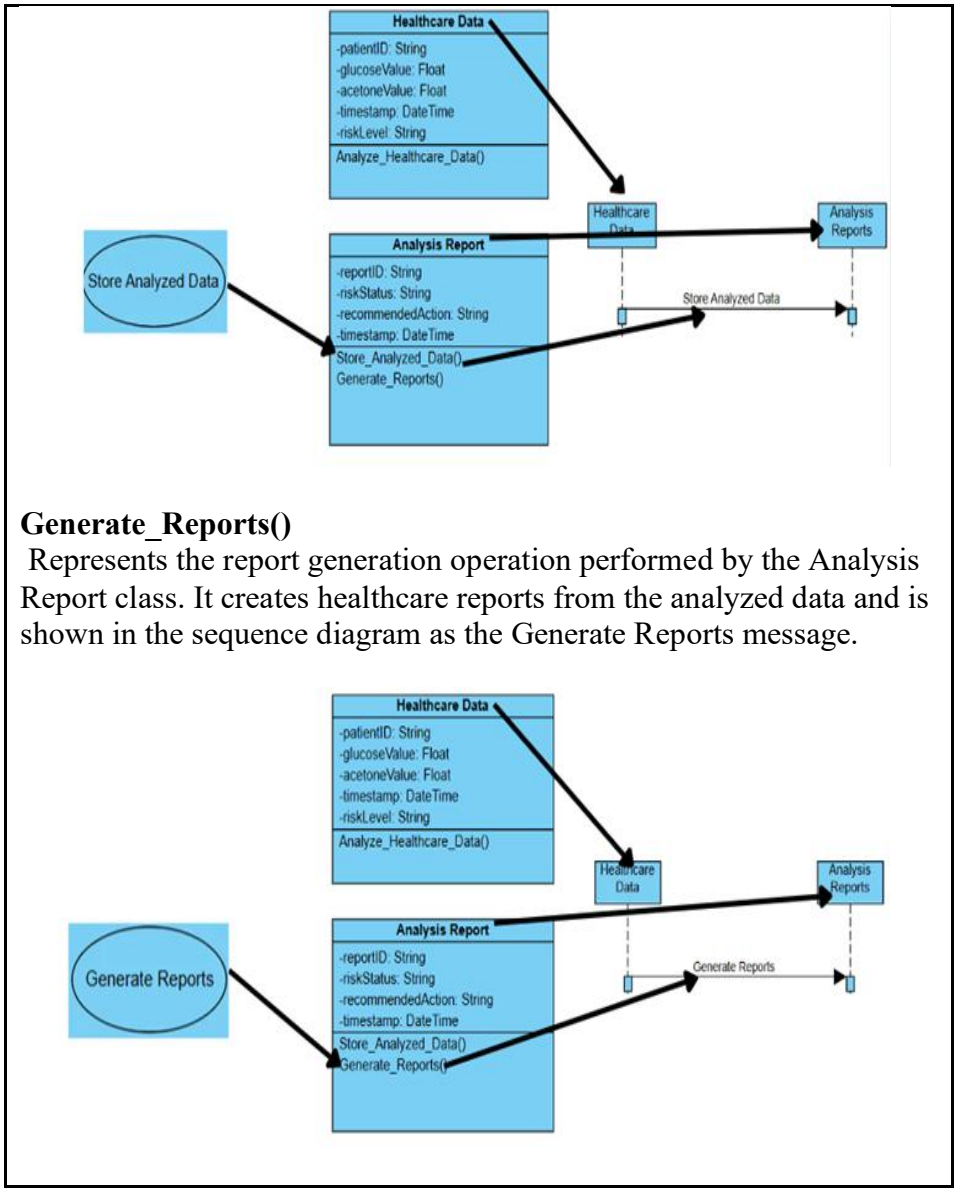
Analyze_Healthcare_Data()

Represents the analysis operation performed by the Healthcare Data class to analyze glucose and acetone readings and determine the patient's metabolic risk level.



Store_Analyzed_Data()

Represents the storage operation performed by the Analysis Report class. It stores the analyzed healthcare data and is shown in the sequence diagram as the Store Analyzed Data message.



Generate_Reports()

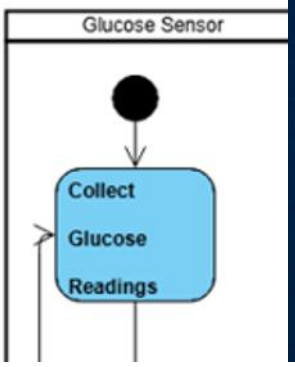
Represents the report generation operation performed by the Analysis Report class. It creates healthcare reports from the analyzed data and is shown in the sequence diagram as the Generate Reports message.

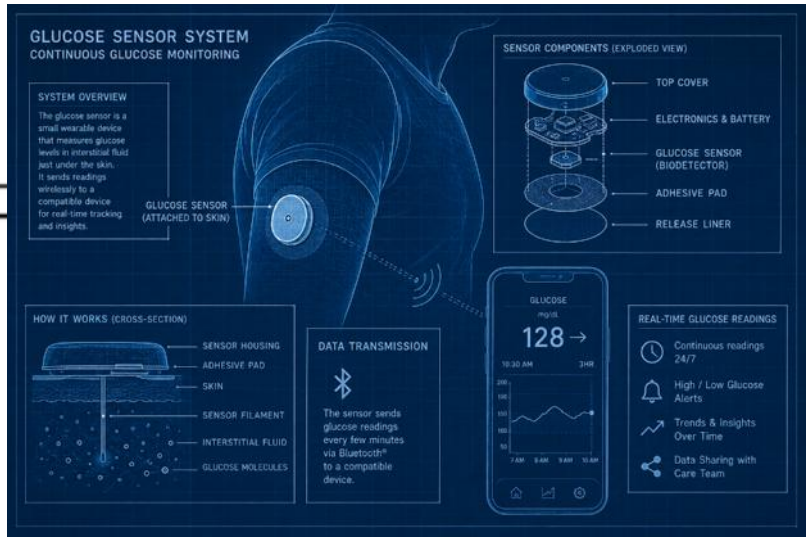
8. Screens Dialogue

8.1. Screens Format Design with Reference to Activity Diagram

- **Ali NagyPart**

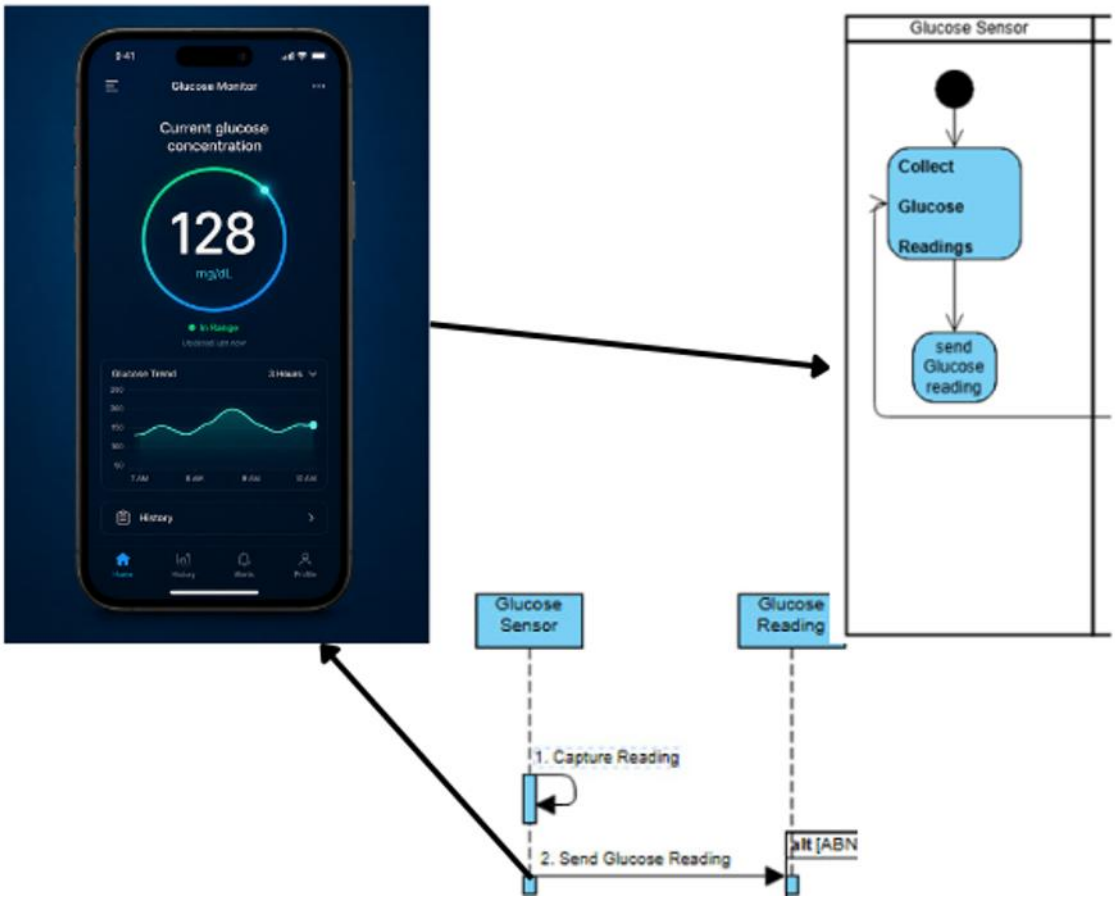
1: Collect Glucose Readings: The Glucose sensor starts working and collecting the glucose levels in the blood of the pateint





The Glucose sensor Starts monitoring the blood glucose level here

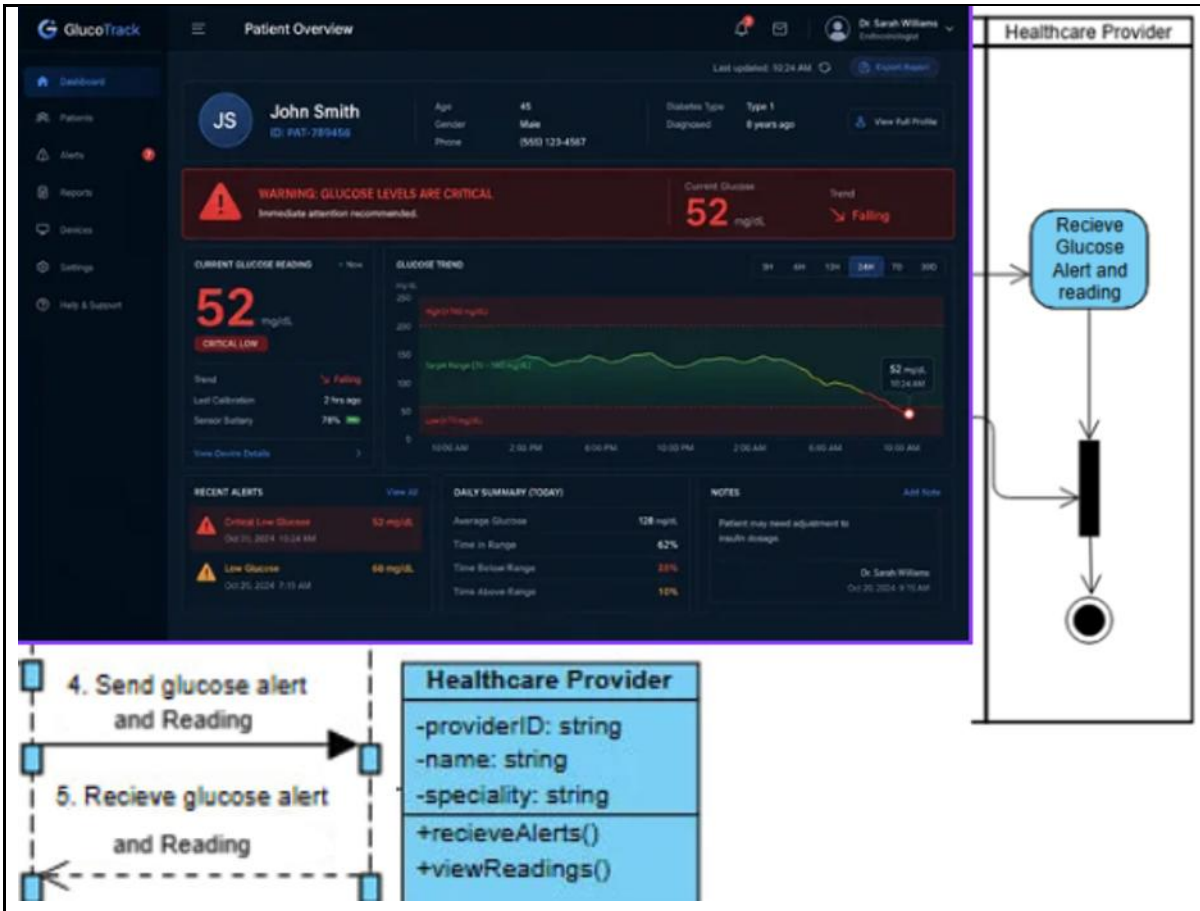
2: Send Glucose Readings:



That Screen is displayed inside the application with the varying glucose readings that are being captured continuously from the glucose sensor, sending the readings to the healthcare monitoring system, making the glucose readings appear in the application of the patient.

3: Sending the Glucose Alert and receiving it:

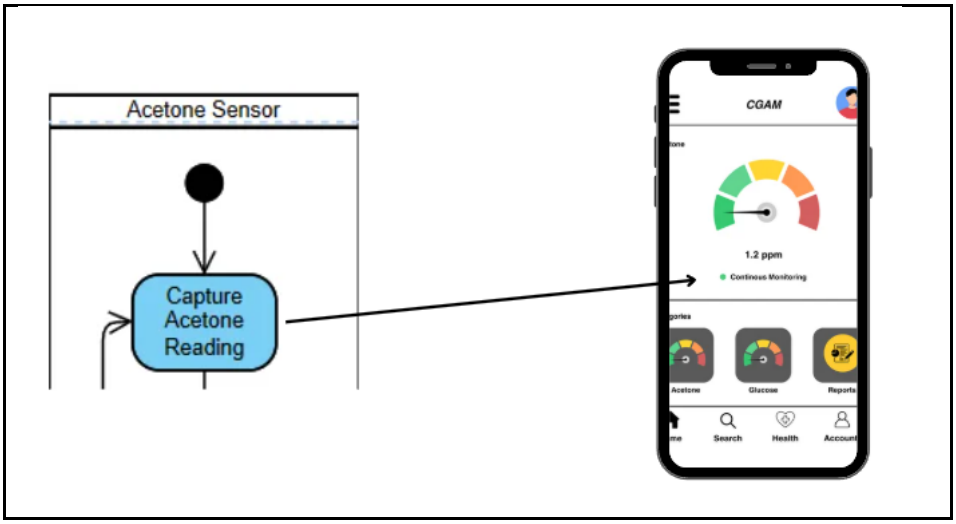

Verizon
Monday, May 20
9:41
Warning!
now
Glucose Levels Are Critical
Current reading:
52 mg/dL
Swipe up to unlock
Alert
Patient
Healthcare Provider
Create and Send Alert and glucose reading
Recieve Glucose Alert and reading
Recieve Glucose Alert and reading
[ELS]
4. Send glucose alert and Reading
5. Recieve glucose alert and Reading



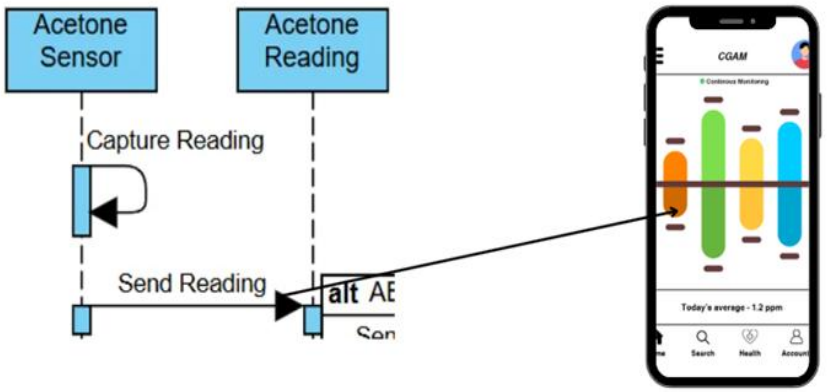
This represents the Alert Sent so the healthcare provider through the dashboard and to the patient through the mobile application the healthcare monitoring system.

• **Nour Marai Part**

Monitoring Screen
- *Capture Acetone Reading* This screen displays the current acetone concentration level and monitoring status during the healthcare monitoring process.



Acetone Readings Screen
- Send Reading - Receive Reading This screen displays the acetone readings transmitted from the acetone biosensor to the healthcare monitoring system.

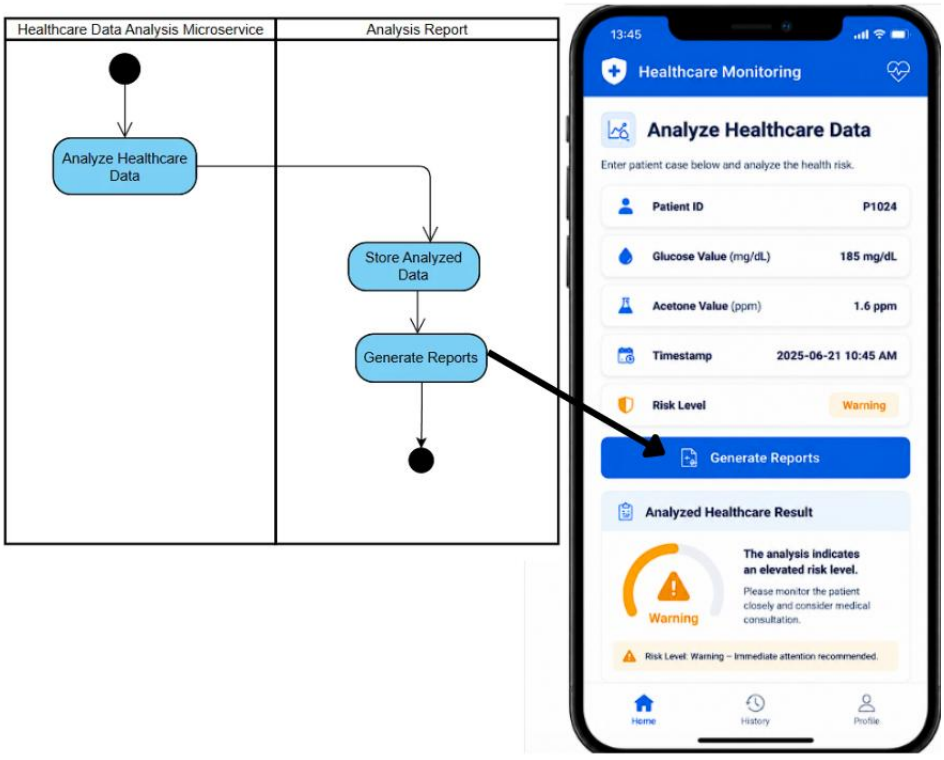


Alert Notification Screen
- Send Alert - Receive Alert

This screen displays healthcare warning notifications whenever abnormal acetone concentration levels are detected.



• Youssef Alaa Part



9. Future Development Work

The further development of the proposed smart health care monitoring system can further enhance the effectiveness, intelligence and scalability of diabetes healthcare management. The current system is effective in integrating continuous glucose monitoring and wearable acetone skin biosensor technology for early detection of diabetic ketoacidosis (DKA) and real-time healthcare monitoring, but there are still several future enhancements that can further enhance the healthcare capabilities of the system and improve overall patient care.

AI and machine learning technologies are expected to be one of the most significant future advancements. Future systems could also incorporate predictive analytical algorithms that are able to detect abnormal metabolic patterns earlier and predict the likelihood of DKA before dangerous symptoms are present. This would help with better clinical decision making and better preventive health care intervention. Besides this, intelligent healthcare recommendation systems that give personalized healthcare advice and monitoring support based on the patient's real-time metabolic condition can be integrated.

Wearable acetone skin biosensors is another important technology that needs improvement in the future. Future sensor designs should aim for greater sensor accuracy, sensitivity, stability and long-term reliability, as well as decreased calibration needs and environmental interference. Further development could also help realise more flexible, comfortable and energy-efficient wearable biosensors to enhance the performance of continuous monitoring for patients.

The system could also extend its connectivity to commercial Continuous Glucose Monitoring (CGM) devices, wearable healthcare technologies and Internet of Medical Things (IoMT) platforms. This integration would facilitate the more efficient exchange of healthcare information between the wearable devices, mobile

healthcare applications, cloud healthcare platforms and healthcare institutions. In addition, future health care dashboards could offer more sophisticated representation of metabolic trends, self-service health care reporting, and telemonitoring.

Other development areas may include strengthening cybersecurity in healthcare and the cloud healthcare infrastructure. A more robust healthcare data protection system, including better healthcare data encryption, secure authentication methods, and increased patient privacy features, could be introduced to enhance protection of healthcare data and facilitate secure remote healthcare services.

Moreover, the system can be extended to add emergency healthcare response capabilities, where critical metabolic conditions automatically send emergency notifications to healthcare providers or healthcare institutions. Enhanced patient group-specific accessibility, as well as more intuitive and engaging mobile app interfaces and personalised notifications could also make mobile healthcare applications more accessible.

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